A.1 Kinematics

A.2 Forces and Momentum

The important stuff. Every idea explained from the ground up.

WHAT THIS TOPIC IS ABOUT Kinematics is just the **description of motion**: where is something, how fast is it going, and how quickly is that speed changing? It does **not** ask *why* it moves like that (that is forces, the next topic). Everything in this guide grows out of one chain of ideas: **position → velocity → acceleration**.

How this guide is organised

Chapter	What is in it	How to use it
1. Kinematics (A.1)	Motion described without asking why: displacement, velocity, acceleration, motion graphs, SUVAT, free fall, projectiles and fluid resistance.	Read in order the first time. Come back to single sections when you are stuck.
2. Forces and Momentum (A.2)	Why things move: Newton's three laws, free-body diagrams, the forces you need to know, momentum and impulse, collisions and circular motion.	Read after Chapter 1: it uses the kinematics equations throughout.

The six-step method for any problem

Step	What you do	Why
1. Draw	Sketch the situation. Choose the origin and which direction is **positive**.	Signs are the biggest source of lost marks. Decide them once, at the start.
2. List	Write what you **know** and what you **want**, with units.	Shows you which quantities are missing.
3. Choose	Pick the equation (or graph rule) that contains what you want and **leaves out** the quantity you do not have.	Avoids needless extra calculations.
4. Rearrange	Solve for the unknown **in symbols first**.	Fewer rounding errors; easier to check.
5. Substitute	Put in numbers **with units** and calculate.	Wrong units show up as wrong answers.

Step	What you do	Why
6. Check	Is the sign right? Is the size sensible? Does it match a graph or a limiting case?	Catches slips before the examiner does.

Unless a question says otherwise: take $\mathbf{g = 9.81\ m\ s^{-2}}$ (the value in the IB data booklet) and **ignore air resistance**.

Contents

Section	Title
Chapter 1	**Kinematics**
1.1	The big picture: how everything connects
1.2	Distance, displacement and signs
1.3	Speed and velocity
1.4	Acceleration
1.5	Motion graphs: gradients and areas
1.6	Equations of motion (SUVAT)
1.7	Free fall
1.8	Vectors: splitting into components
1.9	Projectile motion
1.10	Fluid resistance and terminal speed
1.11	Experiments: measuring g and uncertainty
1.12	Relative velocity in a straight line (enrichment)
Chapter 2	**Forces and Momentum**
2.1	Forces and Newton's first law
2.2	Free-body diagrams
2.3	The forces you need to know
2.4	Newton's second law
2.5	Forces at an angle and on slopes
2.6	Newton's third law
2.7	Momentum
2.8	Impulse and force–time graphs
2.9	Conservation of momentum: collisions and explosions
2.10	Circular motion

Chapter 1

Kinematics

1.1 The big picture: how everything connects

Kinematics looks like a long list of topics: graphs, equations, free fall, projectiles. They are all **one idea seen from different angles**. Three quantities describe motion, and each one is *how quickly the previous one changes*:

- **Position** (s) tells you *where* an object is.
- **Velocity** (v) tells you *how quickly position is changing*, and in which direction.
- **Acceleration** (a) tells you *how quickly velocity is changing*.

On a graph, "how quickly something changes" is a **gradient** (slope), and going the other way (adding up all the little changes) is an **area**. That is why graphs are so central: they are the picture of this chain.

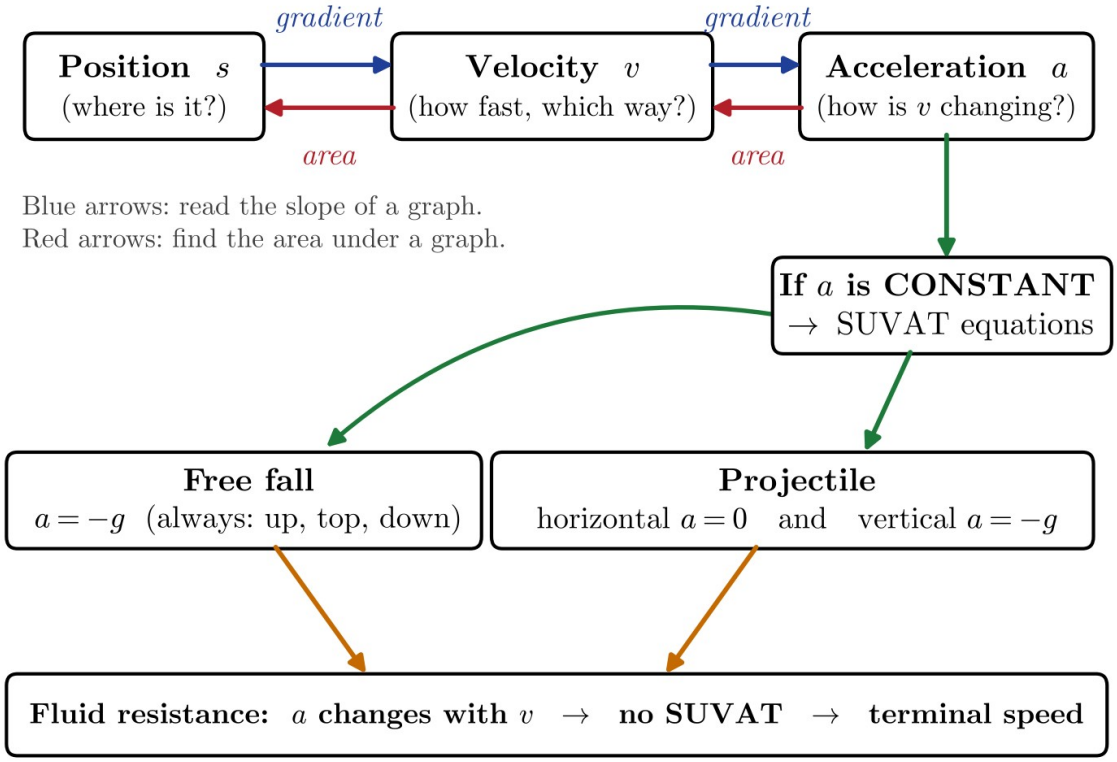


Figure 1.1: How the ideas in kinematics connect. Gradients take you along the chain from position to velocity to acceleration; areas take you back. Constant acceleration gives SUVAT; free fall and projectiles are special cases; fluid resistance breaks the "constant acceleration" assumption.

IN PLAIN ENGLISH Velocity answers "how fast is my position changing?" Acceleration answers "how fast is my velocity changing?" If you can say those two sentences without looking, you already understand most of this topic. Everything else is learning to **read** that idea from a graph, an equation or a diagram.

1.2 Distance, displacement and signs

Position is always measured **relative to a chosen origin**. You choose the origin and which direction counts as positive; the physics does not care, but you must stay consistent.

Distance is the total length of the path travelled. It only ever adds up, so it is never negative. **Displacement** is the straight-line change in position from start to finish, *with direction*:

$$\Delta s = s_{\text{final}} - s_{\text{initial}} \tag{1.1}$$

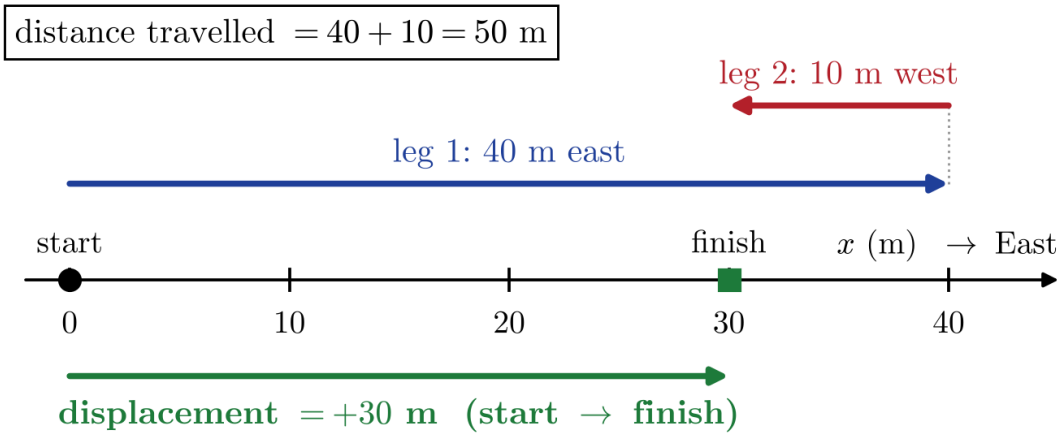


Figure 1.2: A walk of 40 m east followed by 10 m west. The distance travelled is 50 m, but the displacement is only +30 m (30 m east of the start).

Quantities that have only a size are **scalars**. Quantities that also have a direction are **vectors**. In one dimension, direction is shown by a **sign**: after you choose "east is positive", a displacement of −5 m simply means 5 m to the west.

Scalar (size only)	**Vector (size and direction)**	**Link**
distance (m)	displacement (m)	displacement is distance *with direction*, measured start → finish
speed (m s^{-1})	velocity (m s^{-1})	velocity is displacement per unit time
time (s)	acceleration (m s^{-2})	acceleration is change of velocity per unit time

WATCH OUT A **negative** sign means "opposite to the positive direction". It does not mean "smaller" or "wrong". A velocity of −8 m s^{-1} is *faster* than +3 m s^{-1}. Also: the distance travelled is always **at least as big** as the size of the displacement. They are equal only for straight-line motion that never turns back.

1.3 Speed and velocity

Speed is how much distance is covered per second. **Velocity** is how much *displacement* is covered per second, so velocity has a direction. For a whole journey we use averages:

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}} \tag{1.2}$$

$$\text{average velocity } \bar{v} = \frac{\Delta s}{\Delta t} = \frac{s_2 - s_1}{t_2 - t_1} \tag{1.3}$$

An **average** hides what happens in between. The **instantaneous velocity** is the velocity at one moment. It is what you get from the average when you make the time interval Δt shrink towards zero:

$$v = \frac{ds}{dt} \left(\text{the gradient of the tangent to the } s-t \text{ graph} \right) \tag{1.4}$$

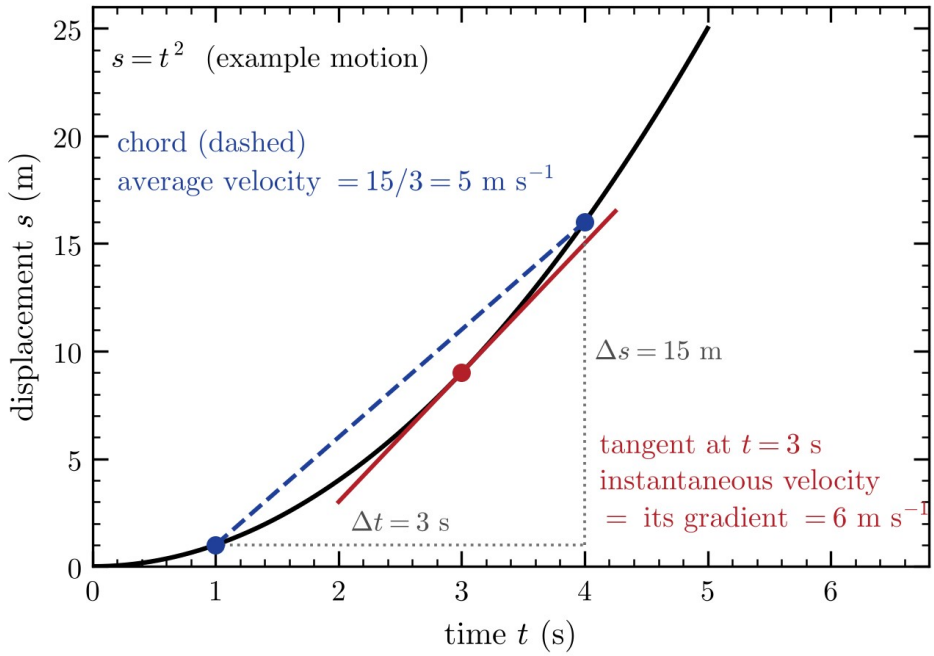


Figure 1.3: On a displacement–time graph, the **chord** between two points (dashed) gives the *average* velocity over that interval. The **tangent** at a single point (red) gives the *instantaneous* velocity at that moment.

	Average velocity	**Instantaneous velocity**
Over what?	a time interval (from t_1 to t_2)	a single instant
From an s–t graph	gradient of the **chord** joining two points	gradient of the **tangent** at that point
Calculation	$\frac{\Delta s}{\Delta t}$	draw tangent, then $\frac{\text{rise}}{\text{run}}$
Speedometer shows...	—	(the size of) the instantaneous velocity

WATCH OUT **Average speed is not the average of two speeds.** If you drive 20 km h^{-1} uphill and 30 km h^{-1} downhill over the same distance d, you spend more *time* at the slow speed. Total time $= \frac{d}{20} + \frac{d}{30} = \frac{d}{12}$, so the average speed is $2d \div \frac{d}{12}$ = **24 km h^{-1}**, not 25. Always use $\frac{\text{total distance}}{\text{total time}}$.

Handy conversion: to go from m s^{-1} to km h^{-1} multiply by **3.6**; to go back, divide by 3.6.

1.4 Acceleration

Acceleration is the rate at which **velocity** changes. If a car's velocity rises from 4 m s^{-1} to 10 m s^{-1} in 3 seconds, it gains 2 m s^{-1} every second, so its acceleration is 2 m s^{-2} (read "2 metres per second, per second").

$$a = \frac{\Delta v}{\Delta t} = \frac{v - u}{t} \tag{1.5}$$

Here **u** is the *initial* velocity and **v** the *final* velocity. Acceleration is a **vector**, so it also has a sign. The key question is not "is a positive or negative?" but **"do v and a point the same way?"**

Velocity v	Acceleration a	What happens to the speed?	Example
+	+	**speeds up** (same direction)	car pulling away to the right
+	−	**slows down** (opposite directions)	car braking while moving right
−	−	**speeds up** (same direction)	ball falling when "up" is positive
−	+	**slows down** (opposite directions)	ball rising when "up" is negative

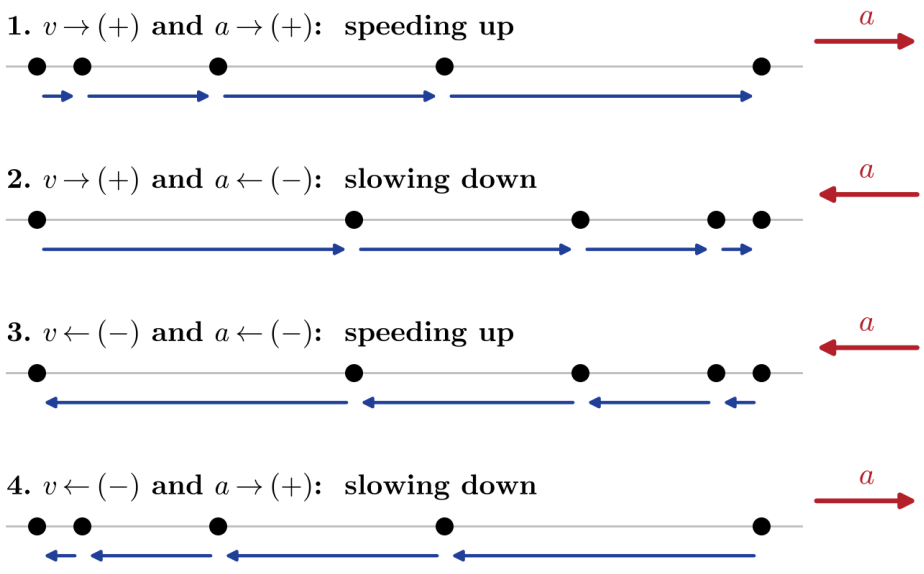


Figure 1.4: Motion diagrams. Each dot is the object's position at equal time intervals; the blue arrows show how far it moved in each interval (its velocity). When the gaps grow the object speeds up; when they shrink it slows down. The red arrow shows the direction of the acceleration.

KEY IDEA **Same direction → speeding up. Opposite directions → slowing down.** The word "deceleration" just means the acceleration points opposite to the velocity. Acceleration can also be non-zero while velocity is momentarily **zero** (for example, at the top of a throw): what matters is that the velocity is *changing*.

1.5 Motion graphs: gradients and areas

There are three graphs for the same motion. Each one is read in two ways: by its **gradient** and by the **area** under it.

Graph	**Gradient (slope) gives...**	**Area under the graph gives...**
Displacement–time (s–t)	**velocity**	(nothing useful)
Velocity–time (v–t)	**acceleration**	**displacement**
Acceleration–time (a–t)	(not needed at SL)	**change in velocity** Δv

The three graphs for one journey are shown in Figure 1.5. The object speeds up (A), cruises (B), then slows, stops and reverses (C). Notice how each graph is the slope of the one above it.

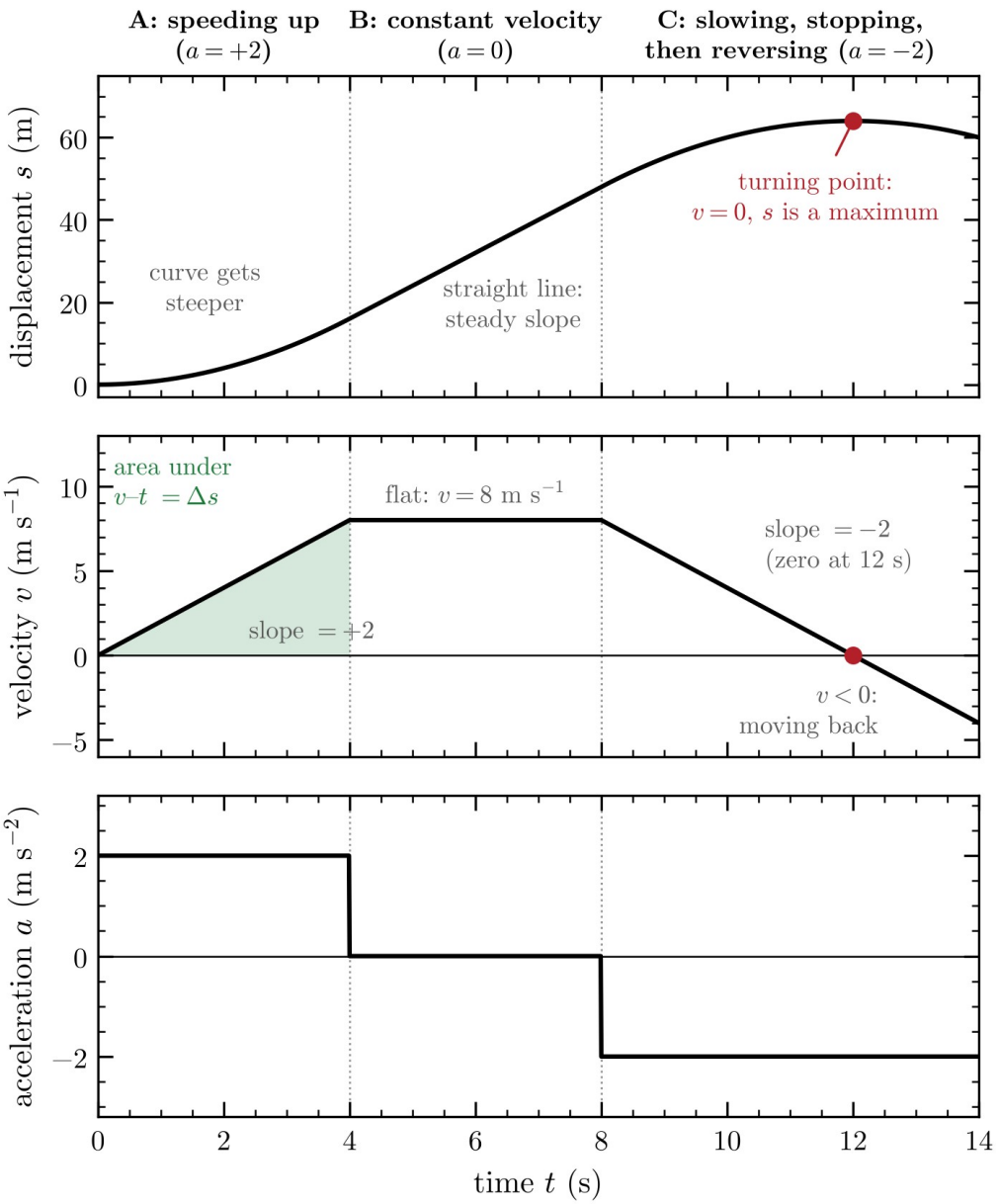


Figure 1.5: The same journey on all three graphs, sharing one time axis. The slope of the s–t graph is the height of the v–t graph; the slope of the v–t graph is the height of the a–t graph. At t = 12 s the velocity is zero, so the displacement is at its maximum: that is the turning point.

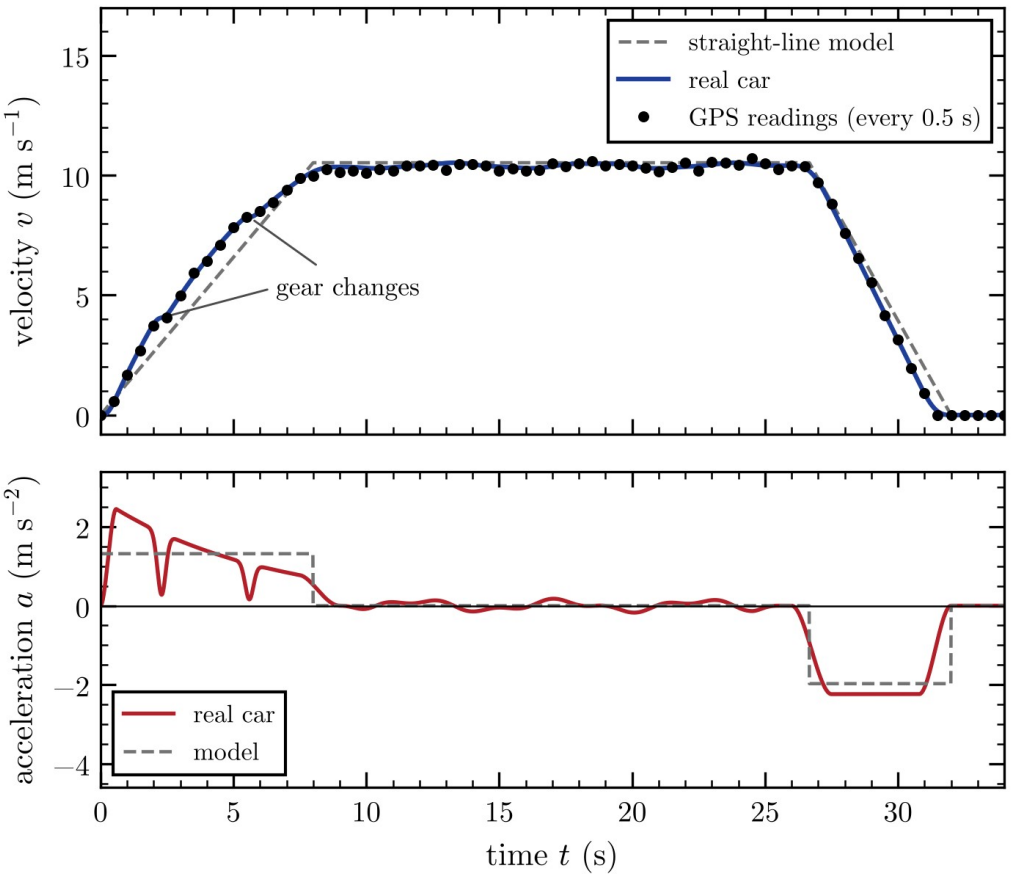


Figure 1.6: A real car pulling away from one set of traffic lights, cruising and braking at the next (computer model, with simulated GPS readings every 0.5 s). A real acceleration cannot jump from one value to another, so the real v–t graph has no sharp corners, and each gear change shows up as a brief dip in a. The straight-line model used in exam questions has the same top speed and the same area under it (about 267 m).

What each shape means

Graph	Shape	Meaning
s–t	horizontal line	at rest (gradient = 0)
s–t	straight sloped line	constant velocity
s–t	curve getting steeper / flatter	speeding up / slowing down
v–t	horizontal line	constant velocity, a = 0
v–t	straight sloped line	constant acceleration
v–t	line crossing the time axis	velocity changes sign: the object **turns round**
a–t	horizontal line (not on the axis)	constant acceleration, so SUVAT works

Using areas on a v–t graph

Areas **above** the time axis are positive displacement; areas **below** it are negative displacement. This gives a two-line rule that comes up in nearly every graph question:

- **Displacement** = the *signed* sum of the areas (add the areas above, subtract the areas below).
- **Distance travelled** = the sum of the *sizes* of all the areas (treat every area as positive).

WATCH OUT (the "hill illusion") **Do not read a graph like a picture of the path.** A v–t graph that slopes downwards does not mean the object is going down a hill; it means its velocity is decreasing. Always say what the *axes* are, then describe what the *line* does. Also: a v–t graph with a negative gradient does not always mean "slowing down": if the velocity is already negative, a negative acceleration makes it speed up in the negative direction (see the table in 1.4).

1.6 Equations of motion (SUVAT)

When the acceleration is **constant**, motion in a straight line can be described exactly by four equations. They are called the SUVAT equations after the five quantities in them:

Symbol	Quantity	Unit
s	displacement (change in position)	m
u	initial velocity (at the start of the interval)	m s^{-1}
v	final velocity (at the end of the interval)	m s^{-1}
a	acceleration (constant)	m s^{-2}
t	time taken	s

Where they come from (no calculus needed)

Take the v–t graph of constant acceleration: a straight line from u at time 0 to v at time t (Figure 1.7). We only need the two graph rules: *gradient = acceleration* and *area = displacement*.

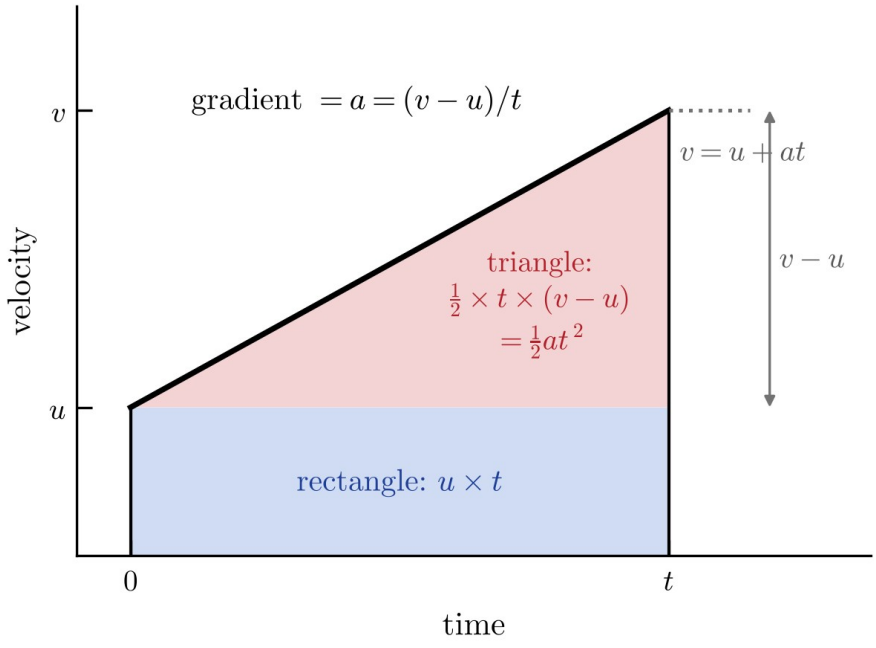


Figure 1.7: A straight-line v–t graph for constant acceleration. The area under it (a rectangle plus a triangle) is the displacement.

- **Gradient.** The gradient is $a = \dfrac{v-u}{t}$. Rearranging gives $v = u + at$.
- **Trapezium.** The area is $\dfrac{1}{2} \times (\text{sum of parallel sides}) \times \text{width} = \dfrac{1}{2}(u+v)t$, so $s = \dfrac{1}{2}(u+v)t$.

- **Rectangle + triangle.** Rectangle: $u \times t$. Triangle: $\frac{1}{2} \times t \times (v-u) = \frac{1}{2} \times t \times at = \frac{1}{2}at^2$. So $s = ut + \frac{1}{2}at^2$.
- **Eliminate t.** Put $t = \frac{v-u}{a}$ into $s = \frac{1}{2}(u+v)t$ to get $s = \frac{v^2-u^2}{2a}$. So $v^2 = u^2 + 2as$.

$$v = u + at \tag{1.6}$$

$$s = \frac{1}{2}(u+v)t \tag{1.7}$$

$$s = ut + \frac{1}{2}at^2 \tag{1.8}$$

$$v^2 = u^2 + 2as \tag{1.9}$$

WHY IT WORKS The middle equation says *displacement = average velocity × time*. For constantly changing velocity the average is exactly halfway between u and v, which is why the $\frac{1}{2}(u+v)$ appears. Every equation here is the same graph, read a different way.

Choosing the right equation

Do not pick the equation that "looks right". Write the five SUVAT quantities as a list, mark the three you **know** and the one you **want**. The fifth quantity is the one you neither know nor want: choose the equation that **leaves it out**.

Equation	Missing variable	Use it when you have no...
$v = u + at$	**s**	displacement
$s = \frac{1}{2}(u+v)t$	**a**	acceleration
$s = ut + \frac{1}{2}at^2$	**v**	final velocity
$v^2 = u^2 + 2as$	**t**	time

WATCH OUT SUVAT works **only** for constant acceleration in a straight line. If the acceleration changes (for example, a skydiver with air resistance), split the motion into stages or use graphs. For a motion in **two stages**, the *final* velocity of stage 1 is the *initial* velocity of stage 2. Quadratic equations in t (from $s = ut + \frac{1}{2}at^2$) give **two** answers: reject any that are physically impossible, such as a negative time.

EXAM TIP For "derive" questions, start by stating the assumption: *constant acceleration*. Then use the trapezium: $v = u + at$ and $s = \frac{1}{2}(u+v)t$ are the two shortest steps to any of the others.

1.7 Free fall

An object is in **free fall** when gravity is the *only* force acting on it. Near the Earth's surface, every object in free fall has the same acceleration, whatever its mass:

$$a = g = 9.81\,\text{m s}^{-2}\,\text{directed downwards} \tag{1.10}$$

"Free fall" does **not** mean "falling". A ball thrown upward is in free fall from the moment it leaves your hand: on the way up, at the top, and on the way down. The acceleration is **always** 9.81 m s^{-2} downward. If you choose *up* as positive, then **a = −9.81 m s^{-2} throughout**, and the SUVAT equations apply.

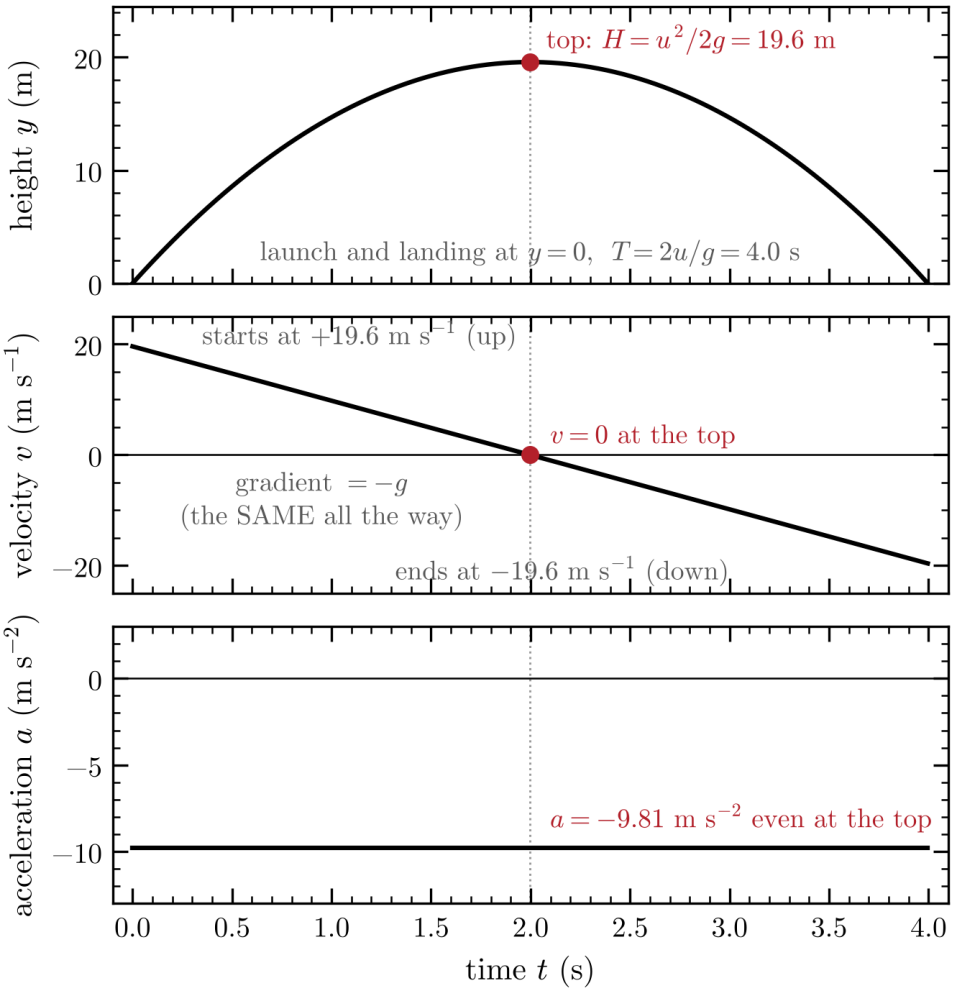


Figure 1.8: A ball thrown upward at 19.6 m s^{-1} (up is positive). The velocity decreases by 9.81 m s^{-1} every second, passes through zero at the top, and keeps going into negative values. The acceleration never changes.

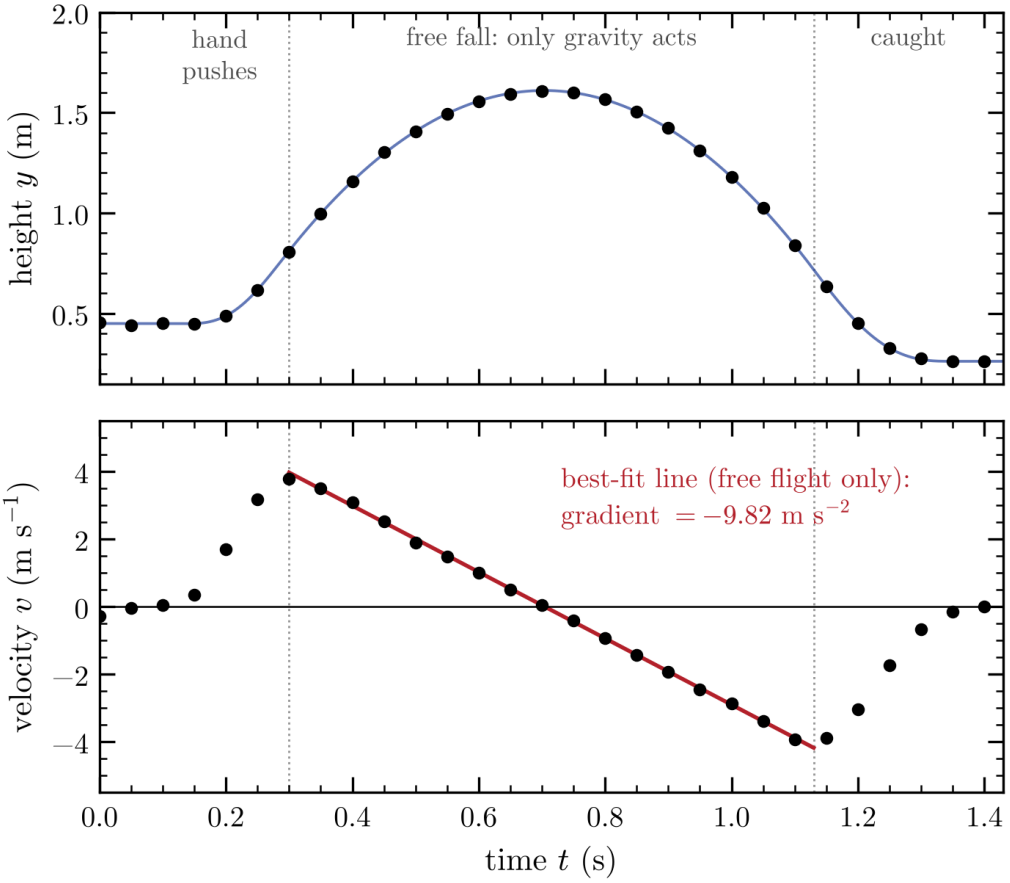


Figure 1.9: Simulated motion-sensor readings (every 0.05 s) for a ball tossed upward and caught. The ball is in free fall only between release and catch: that part of the v–t graph is a straight line with gradient -9.82 m s^{-2}, very close to −g. While the hand is pushing or catching, it is not free fall.

Useful results (launch and landing at the same height, no air resistance)

Quantity	**Result**	**How it comes from SUVAT**
time to the top	$t_{\text{up}} = \frac{u}{g}$	$v = u + at$ with v = 0
maximum height	$H = \frac{u^2}{2g}$	$v^2 = u^2 + 2as$ with v = 0
total time in the air	$T = \frac{2u}{g}$	going up and coming down take equal times
speed on returning to the launch height	= u (but downward)	the motion is symmetric
dropped from rest through height h	$t = \sqrt{\frac{2h}{g}}, v = \sqrt{2gh}$	$s = \frac{1}{2}at^2$ and $v^2 = 2as$ with u = 0

WHY a ≠ 0 AT THE TOP At the very top the velocity is zero **for an instant**, but the acceleration is not zero. If it were, the ball would stay at rest there forever, like a ball with no forces on it. Gravity is still pulling, which is why the velocity goes from positive (rising) to negative (falling) rather than staying at zero.

WATCH OUT The number g is *positive* (9.81). The minus sign belongs to your **sign convention** (up positive means a = −g). If you choose *down* as positive instead, then a = +g. Pick one convention for a problem and use it for *every* quantity: s, u, v and a.

1.8 Vectors: splitting into components

A velocity at an angle can be split into two perpendicular parts that act **independently**: one horizontal, one vertical. This is the tool that turns a curved, two-dimensional motion into two simple one-dimensional motions.

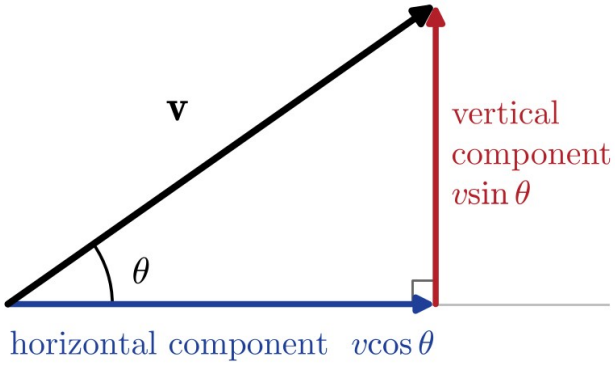


Figure 1.10: Resolving a vector of magnitude v at angle ϑ above the horizontal into a horizontal component v cos ϑ and a vertical component v sin ϑ.

$$v_x = v\cos\theta \tag{1.11}$$

$$v_y = v\sin\theta \tag{1.12}$$

To go back from components to a single vector, use Pythagoras for the size and the tangent for the angle:

$$v = \sqrt{v_x^2 + v_y^2} \tag{1.13}$$

$$\tan\theta = \frac{|v_y|}{|v_x|} \tag{1.14}$$

WATCH OUT **Check which angle you are given.** If ϑ is measured from the *horizontal*: horizontal part = cos ϑ, vertical part = sin ϑ. If ϑ is measured from the *vertical*, they swap. A quick check: when ϑ is small (almost horizontal), the horizontal component should be nearly the whole vector.

Direction of a final answer: use the *sizes* of the components in $\tan\theta = \frac{|v_y|}{|v_x|}$ and then describe it in words, for example "58° below the horizontal". This avoids the calculator giving an angle in the wrong quadrant.

1.9 Projectile motion

A **projectile** is an object that, once launched, moves under gravity alone (ignoring air resistance). Its path is a curve called a **parabola**. Two ideas unlock every projectile problem:

KEY IDEA **1. Horizontal and vertical motion are independent.** Nothing pushes sideways ($a_x = 0$), so the horizontal velocity never changes. Gravity acts only vertically ($a_y = -g$).

2. They share the same clock. The time of flight is the **link** between the two directions.

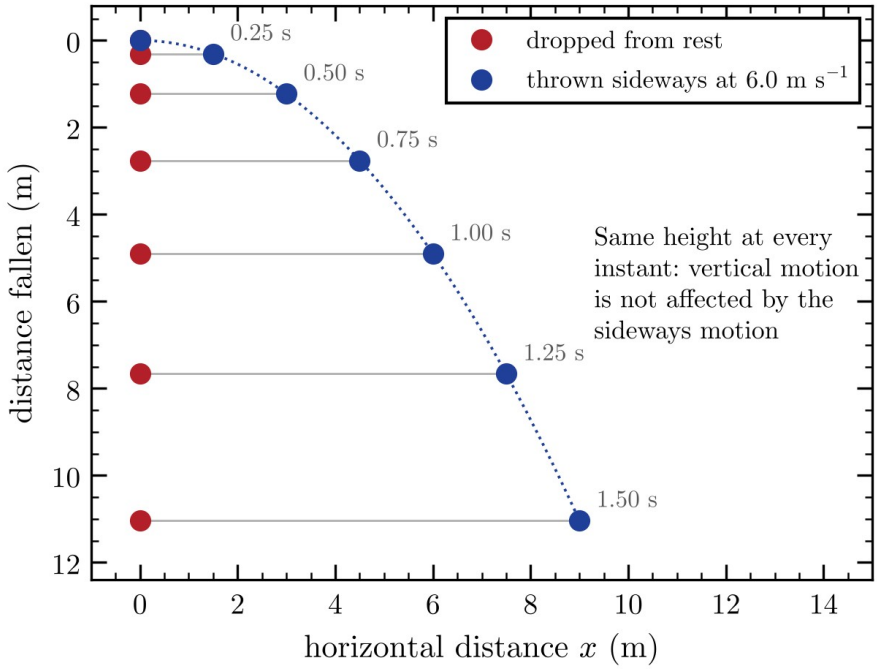


Figure 1.11: One ball is dropped; another is thrown sideways at the same moment. Positions are shown every 0.25 s. At every instant both are at the same height: the sideways motion has no effect on the fall.

	Horizontal (x)	**Vertical (y, up positive)**
Acceleration	0	$-g = -9.81$ m s^{-2}
Initial velocity	$u_x = u\cos\vartheta$	$u_y = u\sin\vartheta$
Velocity at time t	$v_x = u_x$ (constant)	$v_y = u_y - gt$
Displacement at time t	$s_x = u_x t$	$s_y = u_y t - \frac{1}{2}gt^2$
No-time equation	—	$v_y^2 = u_y^2 - 2gs_y$

$$s_x = u_x t \tag{1.15}$$

$$v_y = u_y - gt \tag{1.16}$$

$$s_y = u_y t - \frac{1}{2}gt^2 \tag{1.17}$$

$$v_y^2 = u_y^2 - 2gs_y \tag{1.18}$$

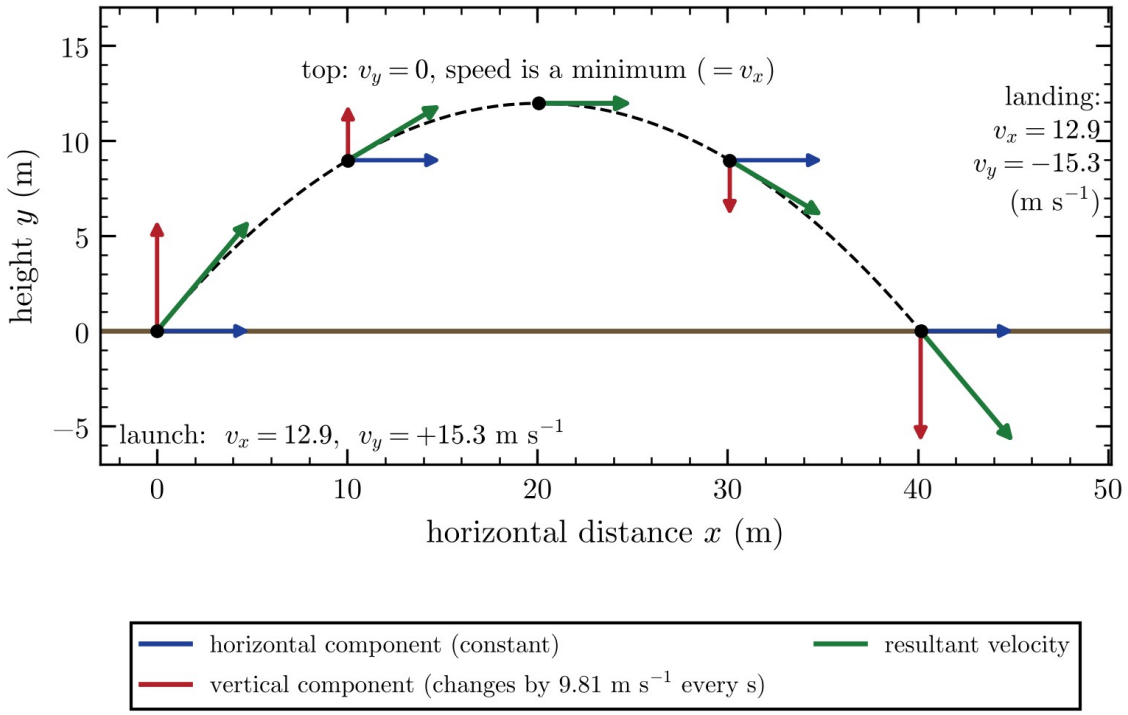


Figure 1.12: Velocity of a projectile launched at 20 m s^{-1}, 50° above the horizontal, shown at equal time steps. The blue horizontal component never changes; the red vertical component shrinks to zero at the top, then grows downward.

What is special about the top of the path?

- The **vertical** velocity is zero ($v_y = 0$), but the object is **not at rest**.
- The **speed is at its smallest**, equal to the horizontal velocity u_x.
- The acceleration is still **g downward**.
- If the object is launched horizontally ($\vartheta = 0$), $u_y = 0$: the starting point *is* the top.

The method for any projectile problem

1. **Resolve** the launch velocity into u_x and u_y.
2. Use the **vertical** equations to find the **time** (from a height, a landing level, or $v_y = 0$).
3. Put that time into the **horizontal** equation to find the distance.
4. Combine v_x and v_y with Pythagoras and $\tan\vartheta$ for the final speed and direction.

Shortcuts when the landing height equals the launch height

If the projectile lands at the *same* height it started from, and there is no air resistance, the flight is symmetric. Setting $s_y = 0$ in Eq. 1.17 and $v_y = 0$ in Eq. 1.16 gives:

$$T = \frac{2u\sin\theta}{g} \tag{1.19}$$

$$H = \frac{(u\sin\theta)^2}{2g} \tag{1.20}$$

$$R = \frac{u^2\sin 2\theta}{g} \tag{1.21}$$

Here T is the time of flight, H the maximum height above the launch point and R the horizontal range ($R = u_x T$, and $2\sin\vartheta\cos\vartheta = \sin 2\vartheta$). Figure 1.13 shows what these mean.

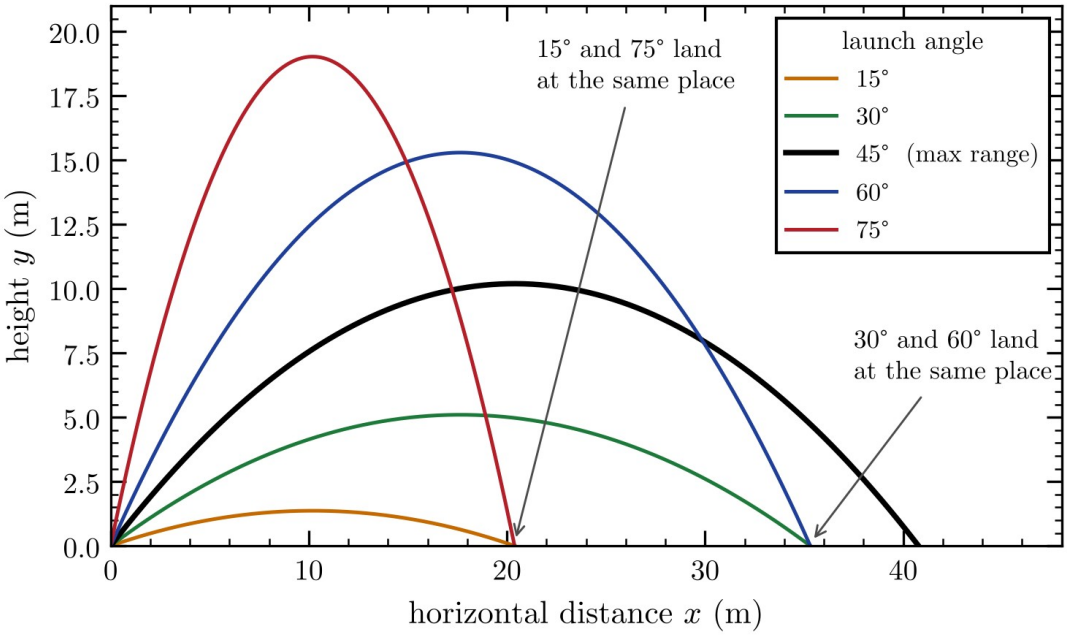


Figure 1.13: Paths for the same launch speed (20 m s^{-1}) at five angles. The range is greatest at 45°. Angles that add to 90° (30° and 60°; 15° and 75°) give the **same** range but different heights and times.

Landing at a different height

If the object lands higher or lower than the launch point, Eqs. 1.19–1.21 **do not apply**. Solve $s_y = u_y t - \frac{1}{2} g t^2$ with the correct s_y (negative if it lands lower). This is a quadratic equation in t: there will be two roots, and you keep the positive, physically sensible one.

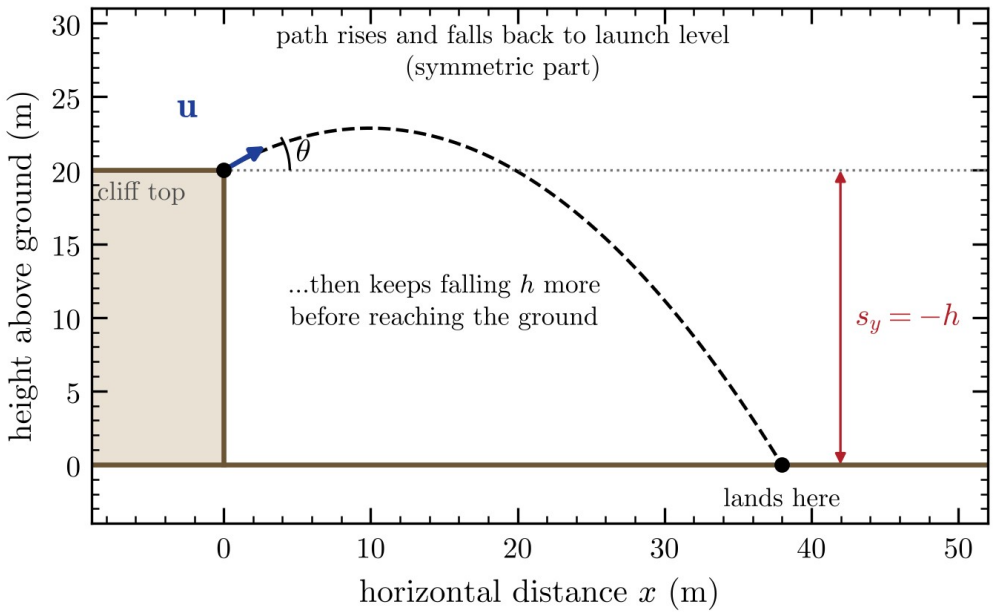


Figure 1.14: A launch from the top of a cliff. The path first rises, then falls back to the launch level, then keeps falling a further height h. Here $s_y = -h$ at the moment of landing.

WATCH OUT The formulas T, H and R are *shortcuts with conditions*: **(i)** same launch and landing height, **(ii)** no air resistance. Using them for a cliff launch, or a ball hit from above the ground, is one of the most common ways to lose marks. When in doubt, go back to Eqs. 1.15–1.18.

EXAM TIP State your assumptions: "neglect air resistance, so $a_x = 0$ and $a_y = -g$." Then show the vertical equation that gives the time **before** you move to the horizontal.

1.10 Fluid resistance and terminal speed

Everything so far ignored air. In reality, anything moving through a **fluid** (air or a liquid) feels a force, called **fluid resistance** or **drag**, that opposes its motion. Drag has four features worth knowing:

- It acts **opposite to the velocity**.
- It **increases with speed**: the faster you go, the harder the fluid pushes back.
- It is larger for a **larger cross-sectional area** and for a less streamlined shape.
- It is larger in a **denser** fluid (water pushes back far more than air).

For an object falling straight down, two forces act: the **weight** mg downward and the **drag** D upward. The resultant force (Newton's second law, which you meet in the next topic) gives the acceleration:

$$a = \frac{mg - D}{m} \tag{1.22}$$

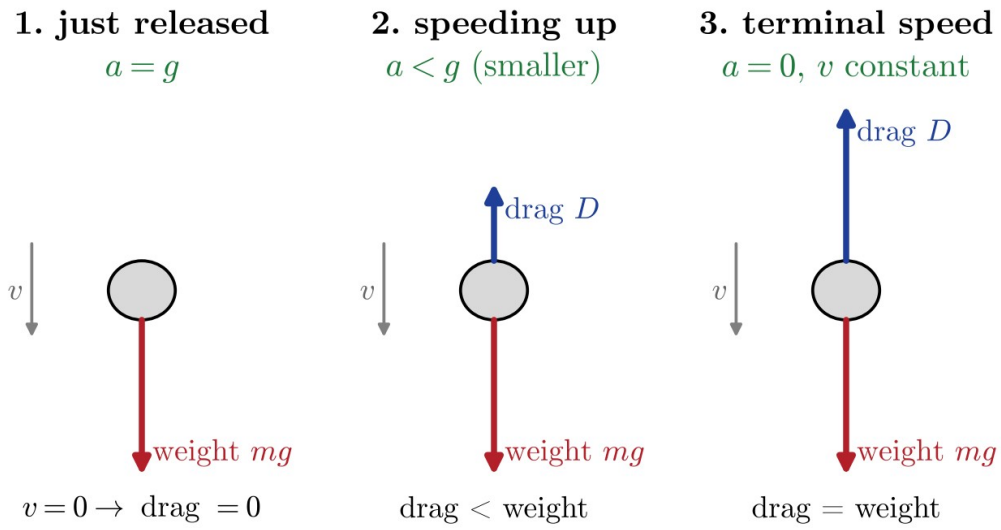


Figure 1.15: Forces on a falling object at three stages. As it speeds up, the upward drag grows until it equals the weight.

Follow the story in Figure 1.15 step by step. This chain of reasoning is exactly what exam questions want to see:

1. Released from rest: speed is zero, so drag is zero. The only force is weight, so **a** = **g**.
2. As the speed increases, drag **increases**. The resultant downward force **decreases**, so the acceleration **decreases** (but the object is still speeding up).
3. Eventually drag **equals** the weight. The resultant force is **zero**, so a = 0 and the velocity is constant. This is the **terminal speed**.

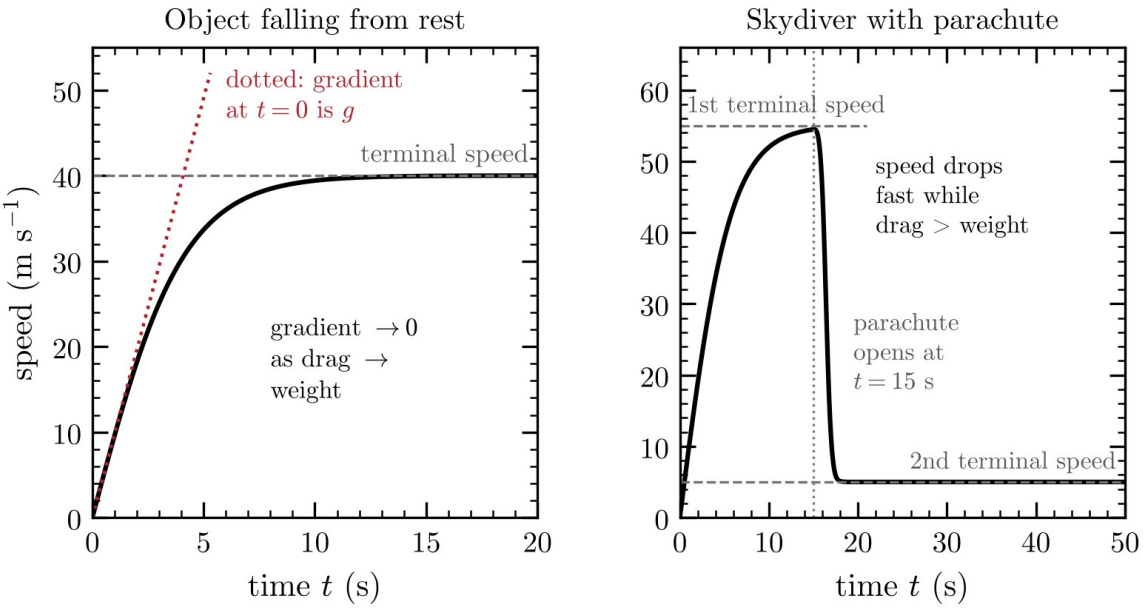


Figure 1.16: Left: speed against time for an object falling from rest; the gradient starts at g and falls to zero as the speed approaches its terminal value. Right: a skydiver. Opening the parachute suddenly makes drag larger than weight, so the skydiver slows down to a new, much lower, terminal speed.

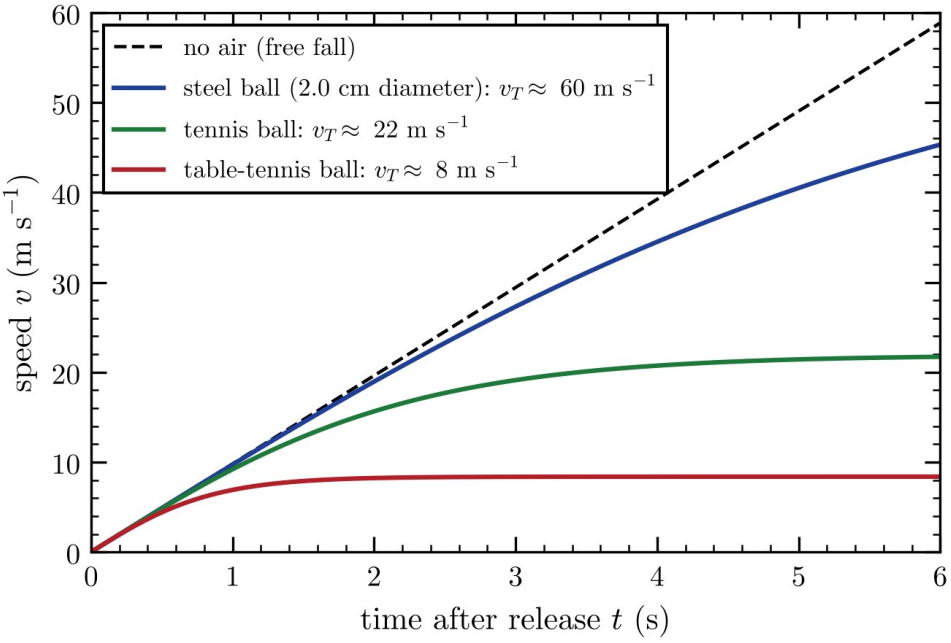


Figure 1.17: Computed speed–time graphs for three balls dropped from rest in air, compared with free fall (no air). The light table-tennis ball reaches its low terminal speed in about a second; the small, dense steel ball is still close to free fall after several seconds.

KEY IDEA **Terminal speed** is the constant speed an object reaches when the drag force equals its weight (resultant force zero, acceleration zero). A parachute has a huge area, so it needs a much *lower* speed before drag can balance the weight.

WATCH OUT Terminal speed does **not** mean the object has stopped accelerating *because gravity switched off*. Gravity is still acting; it is simply balanced by the drag. Also note that the speed-time graph has a gradient which is *decreasing*, not that the speed is decreasing.

Drag on a thrown object

When an object is thrown **upward** through the air, both weight and drag point **down**, so on the way up the acceleration is *greater* than g. At the top the speed is momentarily zero, so drag is zero and the acceleration is exactly g. On the way **down** drag points up, so the acceleration is *less* than g.

Projectiles with fluid resistance (qualitative)

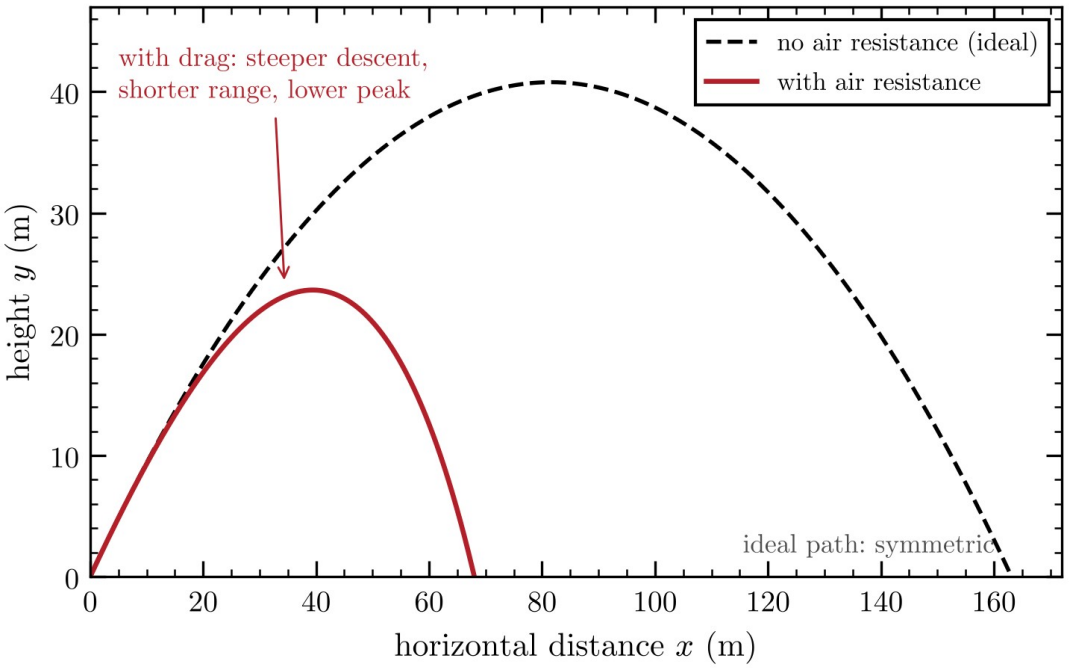


Figure 1.18: A projectile launched at 40 m s^{-1} at 45° with and without air resistance (computer model). Drag makes the path lopsided: lower, shorter, and steeper on the way down.

Quantity	**No air resistance**	**With air resistance**
Horizontal velocity	constant	**decreases** all the way (drag acts backwards)
Vertical acceleration	always $-$g	larger than g on the way up; exactly g at the top; smaller than g on the way down
Maximum height	H	**lower**
Range	R	**shorter**
Time of flight	$\dfrac{2u\sin\theta}{g}$	usually **shorter**; the descent takes longer than the ascent
Shape of path	symmetric parabola	**asymmetric**, steeper on the way down
Landing angle	same as launch angle	**steeper** than the launch angle
Landing speed	equal to launch speed	**smaller** than launch speed

1.11 Experiments: measuring g and uncertainty

A classic experiment measures g by dropping a small, dense object from different heights and timing the fall. Starting from rest, $s = \frac{1}{2} g t^2$ (Eq. 1.8 with u = 0 and a = g). This is not a straight line in s against t, but it **becomes** one if you plot s against $\mathbf{t^2}$:

$$s = \frac{g}{2} t^2 \text{ so } g = 2 \times (\text{gradient of the } s - t^2 \text{ graph}) \tag{1.23}$$

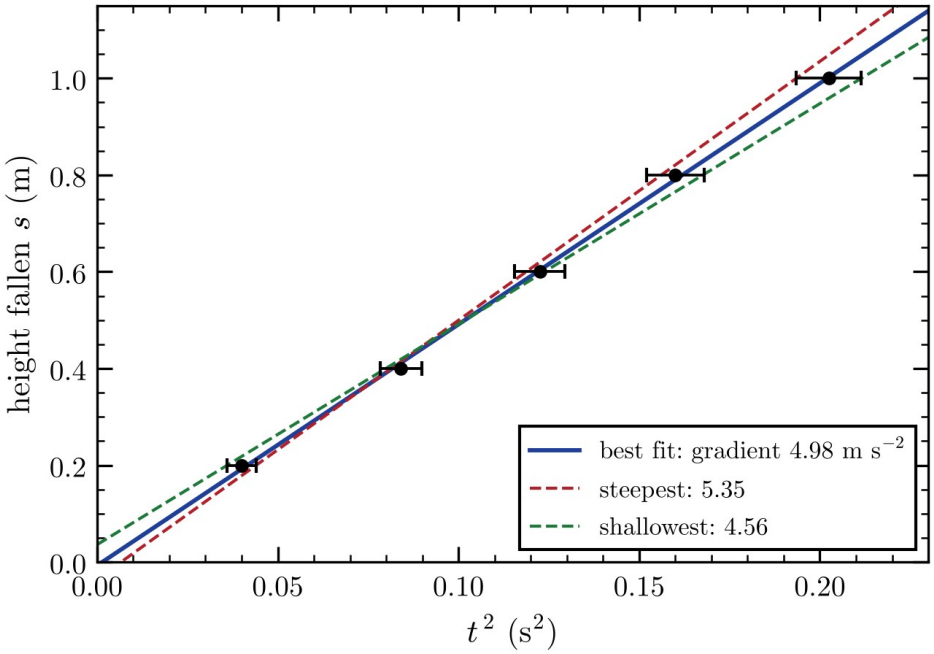


Figure 1.19: Height fallen against t^2 for a dropped ball, with error bars in t^2. The best-fit line (blue) and the steepest and shallowest lines (dashed) that still pass through all the error bars give the uncertainty in the gradient.

Every measurement has an **uncertainty**. Use these rules to combine them:

Operation	**Rule for the uncertainty**	**Example**
a ± b	**add the absolute uncertainties**	(5.0 ± 0.1) + (3.0 ± 0.1) = 8.0 ± 0.2
a × b or a ÷ b	**add the percentage (fractional) uncertainties**	10 ± 2% and 5 ± 3% → product ± 5%
a^n	**multiply the percentage uncertainty by n**	t = 0.50 ± 2% → t^2 has ± 4%
Gradient of a best-fit line	$\frac{\text{steepest} - \text{shallowest}}{2}$	$\frac{5.35 - 4.56}{2} \approx 0.4$

Type of error	**Cause (example)**	**Effect and fix**
Random	reaction time when using a stopwatch; scatter in repeat readings	scatter of points; **repeat and average**

Type of error	Cause (example)	Effect and fix
Systematic	zero error on the ruler; air resistance on the falling ball; a timer that starts late	all values shifted the same way; **cannot** be averaged away; **improve the method**

EXAM TIP Report the final answer with a sensible number of significant figures and its uncertainty, e.g. $\mathbf{g = 9.6 \pm 0.5\ m\ s^{-2}}$. The uncertainty is normally given to **one** significant figure, and the answer to the **same decimal place**.

1.12 Relative velocity in a straight line (enrichment)

Velocity always depends on who is measuring. A passenger walking at 1 m s^{-1} along a train moving at 20 m s^{-1} has a velocity of 21 m s^{-1} *relative to the ground* but 1 m s^{-1} relative to the train. For motion along a line (adding signed velocities):

$$v_{\text{A relative to C}} = v_{\text{A relative to B}} + v_{\text{B relative to C}} \tag{1.24}$$

A common case is a boat in a river. The boat's speed relative to the **water** is fixed by its engine; the water moves relative to the **ground**; the ground-based observer sees the sum.

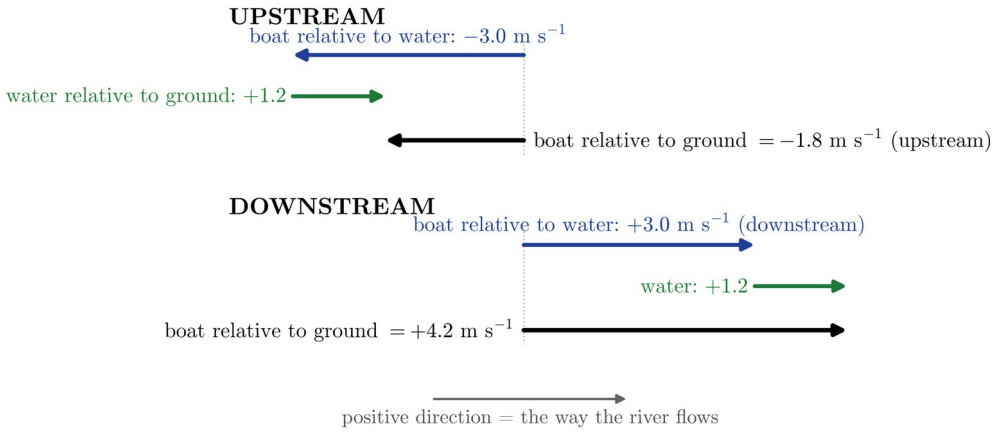


Figure 1.20: A boat that can move at 3.0 m s^{-1} relative to the water, in a river flowing at 1.2 m s^{-1}. Upstream the velocities partly cancel; downstream they add.

EXAM TIP For relative-velocity questions, choose a positive direction first, label each velocity with *"of what, relative to what"* (for example $v_{\text{boat/water}}$, $v_{\text{water/ground}}$) and then simply add. Check that the "inner" labels cancel: boat/water + water/ground = boat/ground.

Chapter 2

Forces and Momentum

WHAT THIS TOPIC IS ABOUT Kinematics described motion without asking why it happens. This topic gives the reason: **forces**. A force changes the velocity of an object, and the rule that links the two is Newton's second law. The same idea, written in terms of **momentum**, explains collisions, explosions, impacts and even why things move in circles.

2.1 Forces and Newton's first law

A **force** is a push or a pull that one object exerts on another. Every force is an **interaction** between two objects: something pushes or pulls *on* something else. Force is a vector, so it has a size and a direction, and its SI unit is the **newton** (N). One newton is the force that gives a 1 kg mass an acceleration of 1 m s^{-2}.

Usually several forces act on an object at once. What matters is their vector sum, called the **resultant force** (or net force). If the forces cancel out, the resultant force is zero.

Newton's first law says what happens when the resultant force is zero:

KEY IDEA **Newton's first law.** An object stays at rest, or keeps moving at a constant velocity (constant speed in a straight line), unless a resultant force acts on it.

This was a revolutionary idea. Everyday experience suggests that things stop unless you keep pushing them, but that is only because friction and drag are always acting. In space, a moving probe keeps going for ever with its engines switched off.

The tendency of an object to keep doing what it is already doing is called **inertia**. The more mass an object has, the more inertia it has: it is harder to start it moving, stop it or turn it.

An object with zero resultant force is in **equilibrium**. There are two kinds:

- **Static equilibrium**: at rest, for example a book on a table.
- **Dynamic equilibrium**: moving at constant velocity, for example a car cruising at a steady 25 m s^{-1} on a straight road, or a skydiver at terminal speed.

WATCH OUT "Constant velocity" needs **no** resultant force. A car cruising at a steady speed still needs its engine, but only to balance drag and friction: the driving force equals the resistive forces, so the resultant is zero. A resultant force is needed only to *change* the velocity.

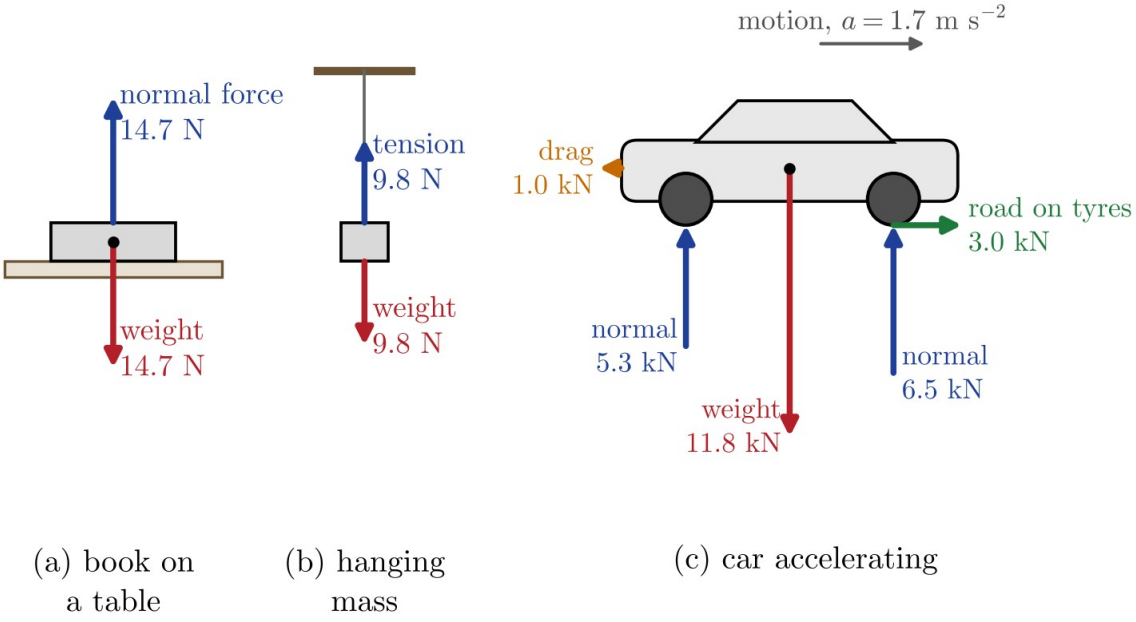
2.2 Free-body diagrams

A **free-body diagram** shows one object on its own, with an arrow for every force that acts **on** it. It is the most important tool in this topic: almost every force problem starts with one.

The method for any free-body diagram

1. Draw the object on its own, as a box or a dot. Do not draw the things around it.
2. Add the weight, acting downwards from the centre of mass.

3. Go round the object: wherever it touches something (a surface, a rope, a fluid), add the force that thing exerts on it.
4. Label every arrow with its name or symbol, and make the lengths roughly match the sizes of the forces.
5. Check: only forces **on** this object, and every force has a cause you can name.



(a) book on a table (b) hanging mass (c) car accelerating

Figure 2.1: Free-body diagrams drawn to scale. (a) A 1.5 kg book resting on a table: the normal force balances the weight. (b) A 1.0 kg mass hanging from a string: the tension balances the weight. (c) A 1200 kg car accelerating: the road pushes the drive wheels forwards with 3.0 kN, drag is 1.0 kN, so the resultant is 2.0 kN forwards.

EXAM TIP Never draw "the force of motion" or "the acceleration" on a free-body diagram. A moving object does not carry a force with it. Only draw forces that something else exerts on the object: weight, normal force, tension, friction, drag, buoyancy, thrust.

2.3 The forces you need to know

Weight

The **weight** of an object is the gravitational force that the Earth exerts on it. Near the Earth's surface it is

$$F_g = mg \tag{2.1}$$

where m is the mass and $g = 9.81\,\text{N kg}^{-1}$ is the **gravitational field strength**. This is the same number as the free-fall acceleration in Chapter 1 (1 N kg^{-1} = 1 m s^{-2}).

WATCH OUT Mass and weight are different. **Mass** (kg) is the amount of matter and does not change from place to place. **Weight** (N) is a force and depends on where you are: a 60 kg astronaut weighs 589 N on Earth but only 97 N on the Moon, where $g = 1.62\,\text{N kg}^{-1}$.

Normal force and tension

When an object presses on a surface, the surface pushes back, at right angles to itself. This is the **normal force** F_N ("normal" means perpendicular). The surface deforms very slightly, like a stiff spring, and pushes back just hard enough to stop the object sinking in. The normal force is *not* always equal to the weight: it is smaller on a slope, and it changes in an accelerating lift (Section 2.4).

A taut string, rope or cable pulls on the objects at each end. This pull is the **tension** T. It always acts along the string, away from the object. For a light string over a smooth pulley, the tension is the same all the way along.

Friction

Friction acts where two surfaces touch, parallel to the surfaces, and opposes their sliding over each other. There are two kinds:

- **Static friction** acts when the surfaces are not sliding. It is only as large as it needs to be to stop sliding, up to a maximum value.
- **Dynamic (kinetic) friction** acts when the surfaces are sliding. It is roughly constant and usually a little smaller than the maximum static friction.

$$F_f \le \mu_s F_N \tag{2.2}$$

$$F_f = \mu_d F_N \tag{2.3}$$

Here μ_s and μ_d are the **coefficients of static and dynamic friction**. They have no units and depend on the two surfaces (rubber on dry tarmac is about 0.8; steel on ice is about 0.03). Notice that friction depends on how hard the surfaces are pressed together (F_N), not on the area of contact.

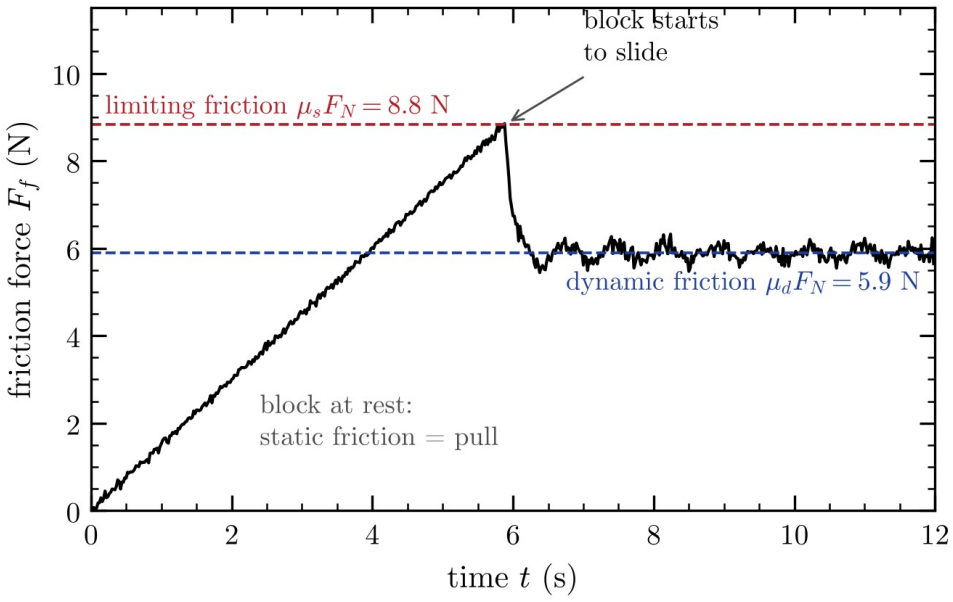


Figure 2.2: Force-sensor reading for a 2.0 kg block pulled with a slowly increasing force (simulated data). While the block is still, static friction exactly matches the pull and rises with it. At the limit, $\mu_s F_N = 8.8$ N, the block starts to slide and the friction drops to the dynamic value, $\mu_d F_N \approx 5.9$ N, with small stick–slip fluctuations.

WATCH OUT $F_f = \mu_s F_N$ is the *maximum* static friction, not the actual friction. A box resting on a level floor with nobody pushing it has **zero** friction on it. If you push it gently with 20 N and it does not move, the static friction is exactly 20 N.

Elastic restoring force (Hooke's law)

A stretched or compressed spring pulls or pushes back towards its natural length. For many springs, the force is proportional to the extension x (the change in length):

$$F_H = -k\,x \tag{2.4}$$

The **spring constant** k (N m^{-1}) measures stiffness: a stiff spring has a large k. The minus sign shows that the force is opposite to the displacement: stretch the spring and it pulls back. Hooke's law only works up to the **limit of proportionality**; beyond it the graph curves.

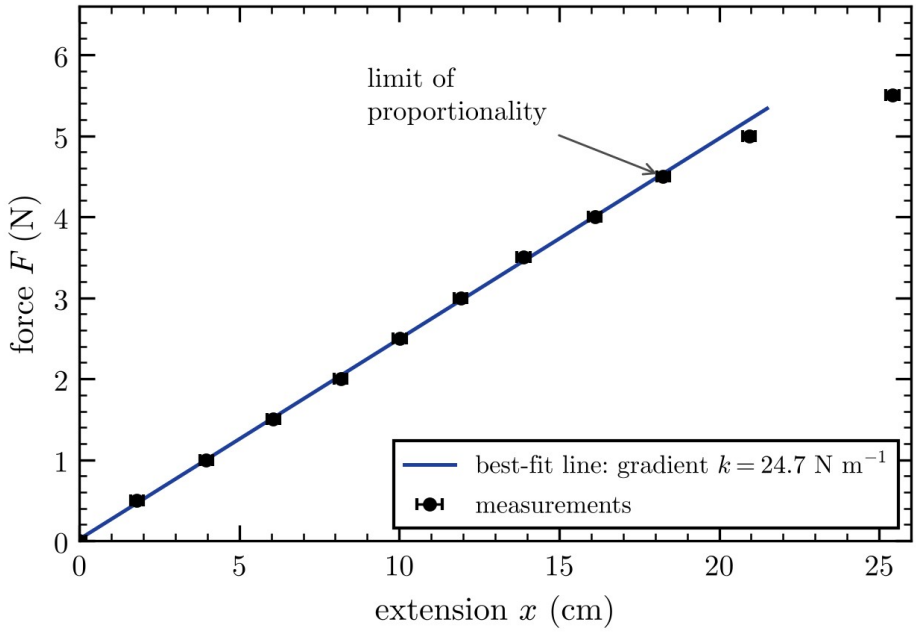


Figure 2.3: Force against extension for a spring (simulated measurements with small random errors). The straight part obeys Hooke's law; its gradient is the spring constant, here about 25 N m^{-1}. Above about 4.5 N the spring passes its limit of proportionality and stretches more than Hooke's law predicts.

Drag and buoyancy in a fluid

An object moving through a fluid feels a **drag force** opposite to its velocity (Section 1.10). For a small sphere moving slowly through a viscous fluid, the drag is given by **Stokes' law**:

$$F_d = 6\pi\eta r v \tag{2.5}$$

where η is the **viscosity** of the fluid (Pa s), r is the radius of the sphere and v is its speed. Stokes' law applies only to smooth spheres moving slowly, with a smooth (non-turbulent) flow of fluid around them.

Any object in a fluid also feels an upward **buoyancy force** (upthrust), equal to the weight of the fluid it pushes aside (**Archimedes' principle**):

$$F_b = \rho V g \tag{2.6}$$

where ρ is the density of the *fluid* and V is the volume of fluid displaced. An object floats when the buoyancy can balance its weight before it is completely submerged.

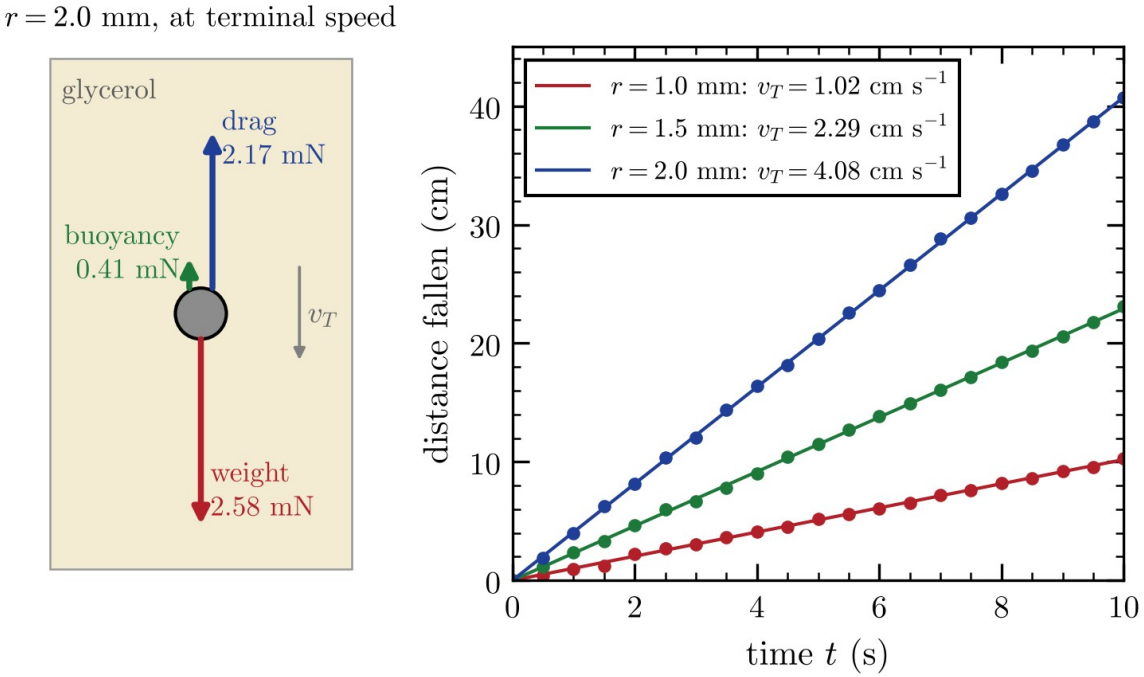


Figure 2.4: Left: forces on a steel ball bearing falling at its terminal speed in glycerol, drawn to scale. Right: position–time data for ball bearings of three sizes (simulated video measurements). Each graph is a straight line, so each ball falls at a constant terminal speed; doubling the radius makes the terminal speed four times larger, as Stokes' law predicts.

At terminal speed the forces balance: $mg = F_b + F_d$. Putting in Stokes' law gives a terminal speed that grows with r^2, which is why fine sand and mud take so long to settle in water.

Force	**Equation**	**Direction**
Weight	$F_g = mg$	towards the centre of the Earth (down)
Normal force	(found from the other forces)	perpendicular to the surface, away from it
Tension	(found from the other forces)	along the string, away from the object
Static friction	$F_f \le \mu_s F_N$	along the surface, opposing the sliding that would happen
Dynamic friction	$F_f = \mu_d F_N$	along the surface, opposing the sliding
Elastic (spring)	$F_H = -kx$	towards the natural length
Viscous drag	$F_d = 6\pi\eta r v$	opposite to the velocity
Buoyancy	$F_b = \rho V g$	upwards

2.4 Newton's second law

When the resultant force on an object is **not** zero, the object accelerates in the direction of the resultant force. The acceleration is proportional to the resultant force and inversely proportional to the mass:

$$F = ma \tag{2.7}$$

Here F is the **resultant** force (the vector sum of all the forces), not any single force. Double the resultant force and the acceleration doubles; double the mass and the acceleration halves.

The method for any $F = ma$ problem

1. Draw a free-body diagram of one object.
2. Choose a positive direction, usually the direction of the acceleration.
3. Write the resultant force as (forces in the positive direction) minus (forces in the negative direction).
4. Set the resultant equal to ma and solve. If you then need a velocity, distance or time, use the SUVAT equations from Chapter 1.

For the car in Figure 2.1(c): resultant force $F = 3.0\,\text{kN} - 1.0\,\text{kN} = 2.0\,\text{kN}$, so $a = \dfrac{F}{m} = \dfrac{2000}{1200} = 1.7\,\text{m s}^{-2}$.

Apparent weight in a lift

Bathroom scales measure the normal force, not your weight. In a lift that accelerates upwards, the floor must push up harder than your weight to give you the upward acceleration. Taking up as positive, $F_N - mg = ma$, so

$$F_N = m(g + a) \tag{2.8}$$

The scale reads more than your weight when the acceleration is upwards (speeding up on the way up, or slowing down on the way down), less when it is downwards, and exactly your weight when the lift moves at constant velocity. In free fall $a = -g$ and the scale reads zero: this is "weightlessness".

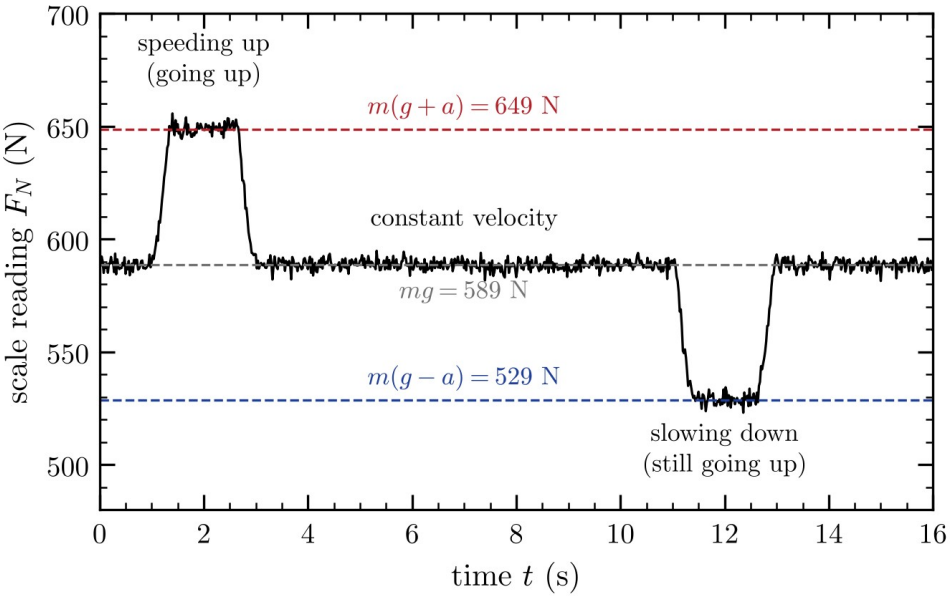


Figure 2.5: Scale reading for a 60 kg person in a lift that starts from rest, rises at constant speed and stops (simulated data, including small vibrations). The reading is 649 N while the lift speeds up at 1.0 m s^{-2}, 589 N (the true weight) at constant velocity, and 529 N while it slows down.

Connected objects

When objects are joined by a string, or one pushes another, they share the same acceleration. You can find it by treating them as **one** object (total mass, external forces only), then draw a separate free-body diagram for one part to find the force between them.

> **WATCH OUT** $F = ma$ uses the *resultant* force. A common mistake is to put the driving force or the tension in place of F and forget the friction or the weight acting the other way. Draw the free-body diagram first, every time.

2.5 Forces at an angle and on slopes

A force at an angle can be replaced by two perpendicular components, exactly as for velocities in Section 1.8. Choose directions that make the problem easy: usually along and perpendicular to the motion.

On a slope at angle θ to the horizontal, resolve the weight **along** and **perpendicular to** the slope:

$$F_{\parallel} = mg\sin\theta \tag{2.9}$$

$$F_{\perp} = mg\cos\theta \tag{2.10}$$

The block does not move into the slope, so the normal force balances the perpendicular component: $F_N = mg\cos\theta$. The component $mg\sin\theta$ pulls the block down the slope; if the block stays still, static friction balances it.

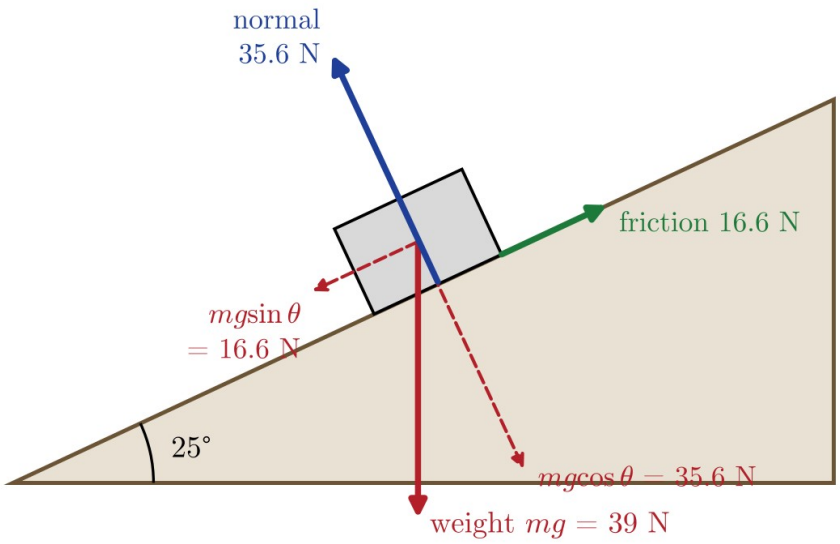


Figure 2.6: A 4.0 kg block at rest on a slope of 25°, drawn to scale. The weight (39 N) is resolved into 16.6 N down the slope and 35.6 N into the slope. The normal force balances the perpendicular part; static friction (16.6 N up the slope) balances the parallel part.

The block is just about to slip when the friction needed equals its maximum value, $mg\sin\theta = \mu_s mg\cos\theta$. The mass cancels, leaving

$$\tan\theta = \mu_s \tag{2.11}$$

So you can measure μ_s by slowly tilting a slope and recording the angle at which the block starts to slide. For the block in Figure 2.6 to stay at rest, μ_s must be at least $\tan 25^\circ = 0.47$.

EXAM TIP On a slope, the angle between the weight and the perpendicular to the slope is the same as the slope angle θ. A quick check: when $\theta = 0$ (flat ground) the component along the slope, $mg\sin\theta$, must be zero, and the component into the slope, $mg\cos\theta$, must be the whole weight.

2.6 Newton's third law

KEY IDEA **Newton's third law.** When object A exerts a force on object B, object B exerts a force on object A that is equal in size and opposite in direction.

The two forces in a third-law pair always:

- act on **different** objects (one on A, one on B);
- are the **same type** of force (both gravitational, both contact, both friction, and so on);
- have **equal size** and **opposite direction**, and act along the same line;
- exist at the same time: you cannot have one without the other.

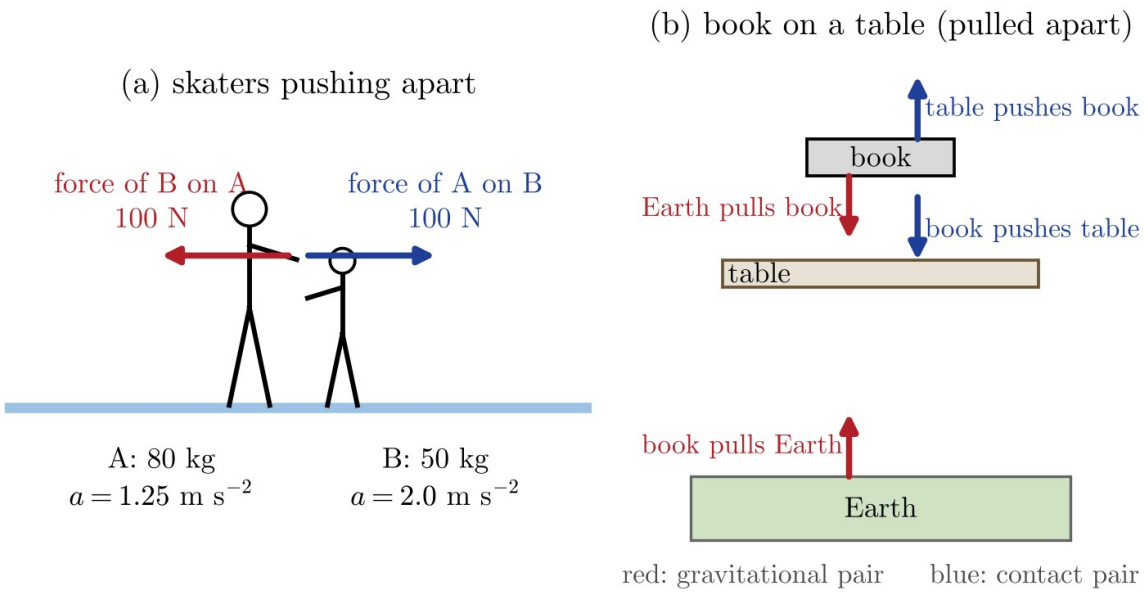


Figure 2.7: Newton's third-law pairs. (a) Two skaters push apart: each feels an equal and opposite force, but the lighter one accelerates more. (b) A book on a table, shown pulled apart for clarity. The weight of the book and the normal force on it act on the **same** object, so they are **not** a third-law pair. Each has its own partner: the book pulls the Earth up, and the book pushes down on the table.

WATCH OUT The weight of a book and the normal force on it are equal and opposite, but this is because of Newton's **first** law (the book is in equilibrium), not the third. They act on the same object and are different types of force. If the table were in an accelerating lift they would not even be equal.

If the forces are equal and opposite, why does anything move? Because the two forces act on different objects, so they never cancel each other. When you walk, you push backwards on the ground with friction, and the ground pushes you forwards with an equal force. The Earth's huge mass means its acceleration is far too small to notice.

2.7 Momentum

The **linear momentum** of an object is its mass multiplied by its velocity:

$$p = m v \tag{2.12}$$

Momentum is a vector, in the same direction as the velocity, with units kg m s^{-1} (which are the same as N s). It measures how hard it is to stop a moving object: a slow lorry and a fast bullet can both have a lot of momentum.

Newton's second law can be written in terms of momentum. The resultant force equals the rate of change of momentum:

$$F = \frac{\Delta p}{\Delta t} \tag{2.13}$$

If the mass is constant, $\Delta p = m \Delta v$, and this becomes $F = m \frac{\Delta v}{\Delta t} = m a$: the same law as before. The momentum form is more general, because it also works when the mass changes, for example a rocket burning fuel or sand falling onto a conveyor belt.

The kinetic energy of an object can also be written in terms of its momentum. Since $E_k = \frac{1}{2} m v^2$ and $v = \frac{p}{m}$,

$$E_k = \frac{p^2}{2m} \tag{2.14}$$

IN PLAIN ENGLISH Force is how quickly momentum changes. A large force changes momentum quickly; a small force needs a long time to make the same change.

2.8 Impulse and force–time graphs

Rearranging Eq. 2.13 gives the **impulse** J, the product of the force and the time for which it acts. The impulse equals the change in momentum it produces:

$$J = F \Delta t = \Delta p \tag{2.15}$$

Impulse has the units N s. In real impacts the force is not constant: it rises from zero to a peak and falls again. The impulse is then the **area under the force–time graph**, and F in Eq. 2.15 is the average force.

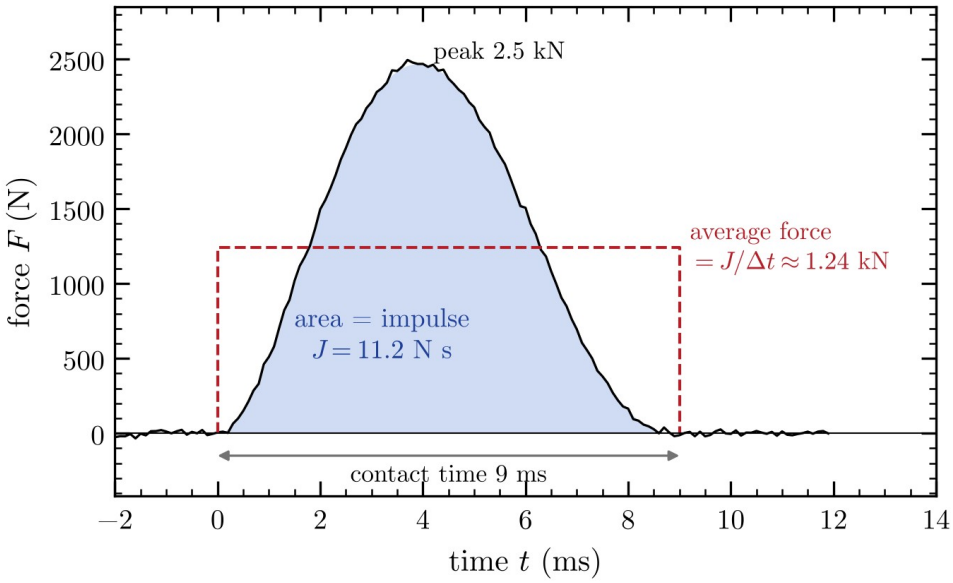


Figure 2.8: Force on a football during a kick, as a force sensor would record it (simulated data, sampled 10 000 times a second). The contact lasts about 9 ms and the force peaks at about 2.5 kN. The shaded area is the impulse, 11.2 N s, which sends the 0.43 kg ball off at 26 m s^{-1}. The dashed rectangle shows the average force: same area, same impulse.

For a given change in momentum, the impulse is fixed. Making the collision last longer therefore makes the average force smaller. This is how many safety features work:

- **Airbags, seatbelts and crumple zones** stretch out the time a passenger takes to stop.
- **Bending your knees** when you land from a jump makes the stopping time longer.
- **Helmets** and **gym mats** contain soft material that squashes slowly.

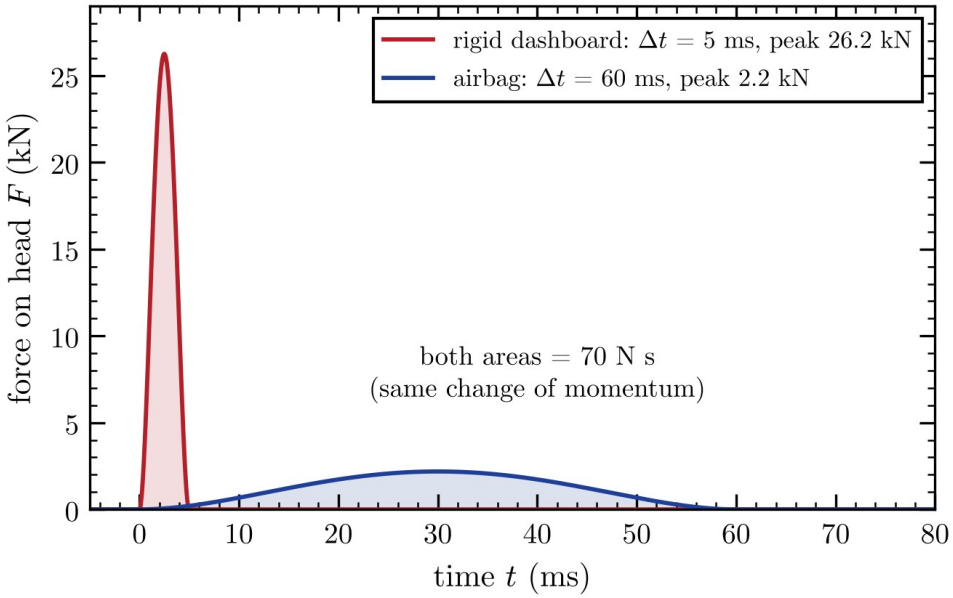


Figure 2.9: The same impulse delivered in two ways (computer model). A 5.0 kg head moving at 14 m s^{-1} (50 km h^{-1}) is brought to rest, so the impulse is 70 N s in both cases. Hitting a rigid dashboard takes about 5 ms and the force peaks at about 26 kN. An airbag stretches the stop to about 60 ms, and the peak force falls to about 2 kN.

WATCH OUT Safety features do **not** reduce the change in momentum or the impulse: the person still stops from the same speed. They reduce the **force** by making the time longer.

2.9 Conservation of momentum: collisions and explosions

When two objects collide, Newton's third law says they exert equal and opposite forces on each other, for exactly the same time. So they receive equal and opposite impulses: whatever momentum one gains, the other loses. This gives one of the most important laws in physics:

> **KEY IDEA** **Conservation of momentum.** In a system on which no external resultant force acts, the total momentum stays constant.

For two objects that collide,

$$m_1u_1+m_2u_2=m_1v_1+m_2v_2 \tag{2.16}$$

where u means a velocity before the collision and v a velocity after it. Velocities are vectors, so choose a positive direction and give velocities in the other direction a minus sign.

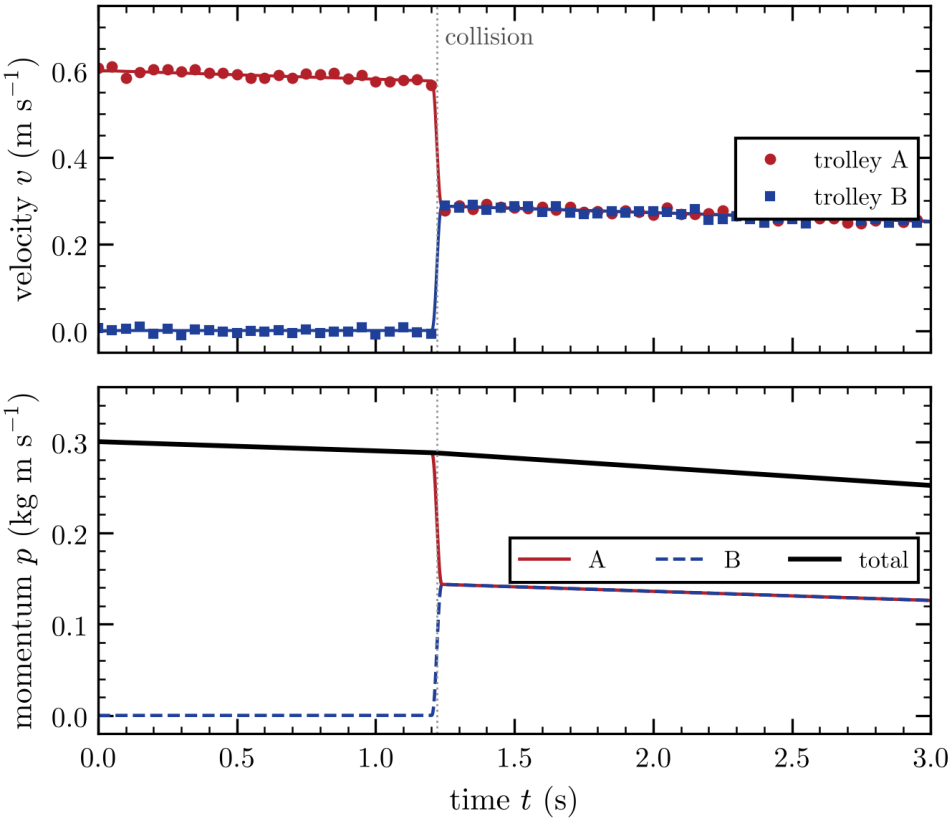


Figure 2.10: Motion-sensor data for two 0.50 kg trolleys (simulated). Trolley A rolls at 0.60 m s^{-1} into trolley B, which is at rest; they stick together and move off at 0.30 m s^{-1}. Bottom: A loses exactly the momentum that B gains, so the total momentum (0.30 kg m s^{-1}) stays the same through the collision. It falls slowly afterwards only because friction, an external force, acts on the trolleys.

Elastic and inelastic collisions

Momentum is always conserved in a collision (if there is no external force). Kinetic energy usually is not: some of it is changed into thermal energy, sound and the work done in permanently bending things.

Type of collision	Momentum	Kinetic energy
Elastic	conserved	conserved (no kinetic energy is lost)
Inelastic	conserved	not conserved (some is changed into other forms)

Type of collision	Momentum	Kinetic energy
Totally inelastic	conserved	the objects stick together: the largest possible loss
Explosion	conserved (often zero before)	increases (comes from stored chemical or elastic energy)

In the trolley collision of Figure 2.10, the kinetic energy before is $\frac{1}{2} \times 0.50 \times 0.60^2 = 0.090$ J and after it is $\frac{1}{2} \times 1.0 \times 0.30^2 = 0.045$ J. Half of the kinetic energy has been changed into other forms, so the collision is totally inelastic, even though momentum is perfectly conserved.

Explosions

In an **explosion** the objects start together, often at rest, so the total momentum is zero before. It must still be zero after: the pieces fly apart with equal and opposite momenta. The kinetic energy comes from a store of energy such as a compressed spring or a chemical reaction. The recoil of a gun and the thrust of a rocket work the same way.

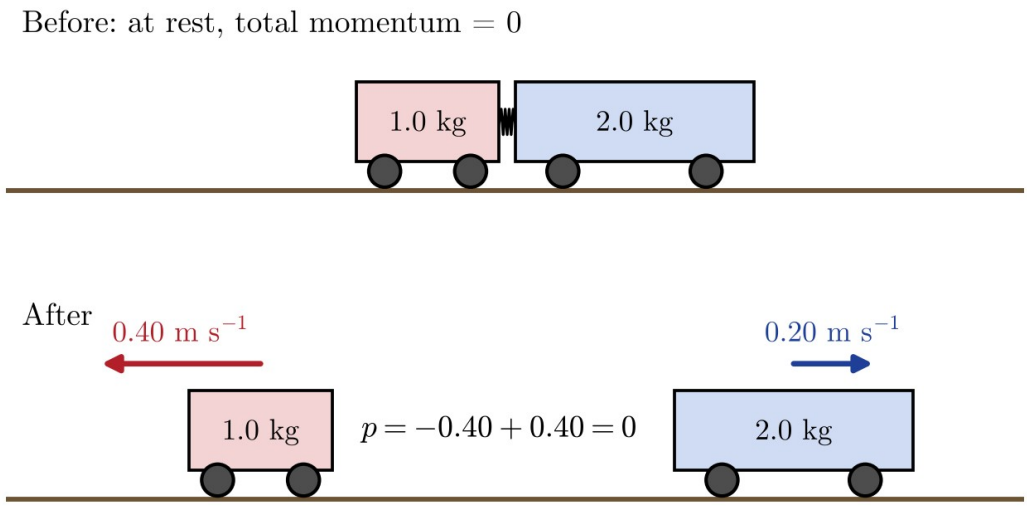


Figure 2.11: Two trolleys at rest are pushed apart by a compressed spring. Afterwards the 1.0 kg trolley moves left at 0.40 m s^{-1} and the 2.0 kg trolley moves right at 0.20 m s^{-1}. Their momenta are equal and opposite (∓ 0.40 kg m s^{-1}), so the total stays zero.

EXAM TIP In any collision question, write "total momentum before = total momentum after", put in the numbers with signs, and solve. Only then think about kinetic energy, which you need if the question asks whether the collision is elastic. Remember that $E_k = \frac{p^2}{2m}$ means that, for the same momentum, the lighter object always has more kinetic energy.

2.10 Circular motion

An object moving in a circle at constant speed is **accelerating**, because the direction of its velocity keeps changing. To describe this motion we use:

- the **period** T: the time for one complete circle;
- the **frequency** f: the number of circles per second, $f = \frac{1}{T}$;
- the **angular velocity** ω: the angle swept out per second, in radians per second (rad s^{-1}).

In one period the object travels a distance $2\pi r$ and sweeps out an angle of 2π radians, so

$$v = \frac{2\pi r}{T} = \omega r \tag{2.17}$$

$$\omega = \frac{2\pi}{T} = 2\pi f \tag{2.18}$$

The velocity is always along the tangent to the circle. The acceleration points towards the **centre** of the circle and is called the **centripetal acceleration**:

$$a = \frac{v^2}{r} = \omega^2 r = \frac{4\pi^2 r}{T^2} \tag{2.19}$$

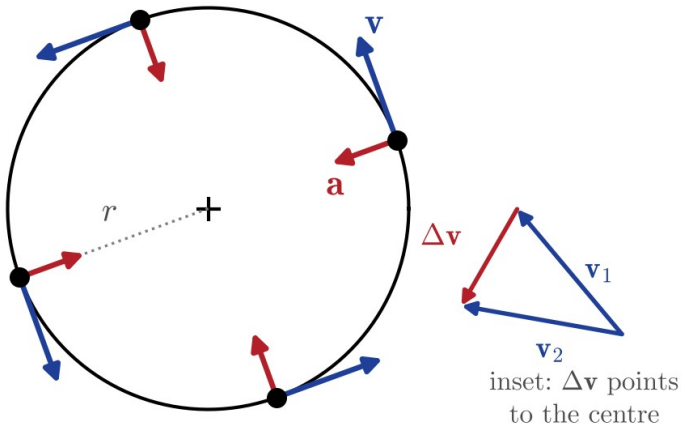


Figure 2.12: An object moving anticlockwise round a circle at constant speed. The velocity (blue) is always tangent to the circle; the acceleration (red) always points to the centre. Inset: in a short time the velocity changes by Δv, which also points towards the centre.

By Newton's second law, a centripetal acceleration needs a resultant force towards the centre, the **centripetal force**:

$$F = \frac{mv^2}{r} = m\omega^2 r \tag{2.20}$$

The centripetal force is not a new kind of force. It is the name for the resultant force towards the centre, and it is always provided by one or more of the real forces from Section 2.3.

Situation	**What provides the centripetal force**
Ball whirled on a string	tension in the string
Car going round a level bend	friction between the tyres and the road

Situation	What provides the centripetal force
Moon orbiting the Earth	gravitational force of the Earth on the Moon
Conical pendulum	horizontal component of the tension
Passenger at the top of a loop-the-loop	weight plus the normal force from the seat (both downwards)

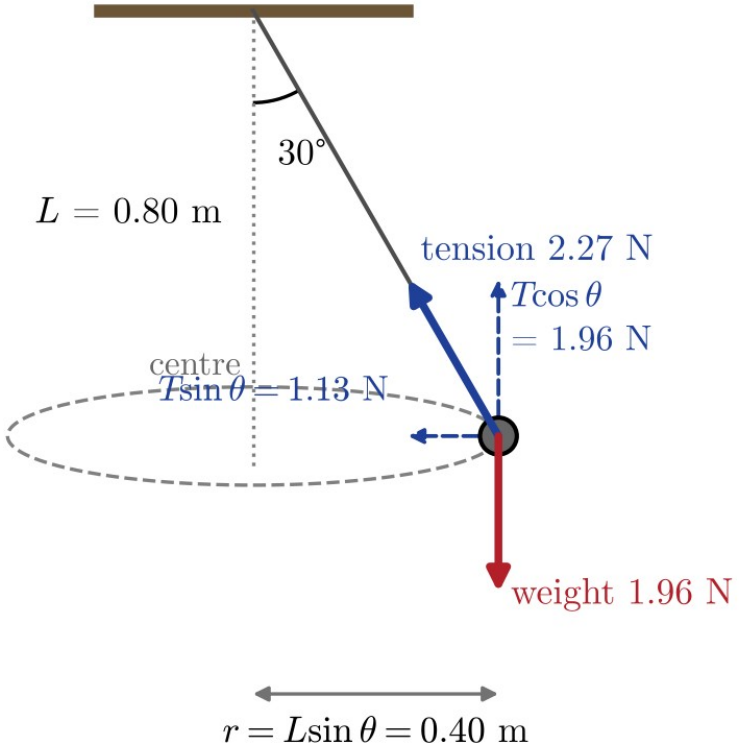


Figure 2.13: A conical pendulum drawn to scale: a 0.20 kg bob on a 0.80 m string moving in a horizontal circle with the string at 30° to the vertical. The vertical component of the tension balances the weight; the horizontal component (1.13 N) is the centripetal force, which gives a speed of 1.50 m s^{-1} and a period of 1.67 s.

For the conical pendulum, $T\cos\theta = mg$ vertically and $T\sin\theta = \frac{mv^2}{r}$ horizontally. Dividing one by the other gives $\tan\theta = \frac{v^2}{rg}$: the faster the bob goes, the further out the string swings.

WATCH OUT There is no outward "centrifugal force" acting on an object moving in a circle. If you feel pushed outwards in a turning car, it is your body trying to keep going in a straight line (Newton's first law) while the car turns underneath you. On a free-body diagram, the resultant force points **inwards**, towards the centre.

EXAM TIP Circular motion questions: (1) draw the free-body diagram; (2) find the resultant force towards the centre; (3) set it equal to $\frac{mv^2}{r}$. Convert any period or frequency into v or ω first, and remember that r is the radius of the circle, which for a conical pendulum is $L\sin\theta$, not the string length L.